Housing Wealth and Labor Supply: Evidence from Geographically-linked Microdata

by

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Abstract

This study examines the causal effect of housing wealth on labor supply using restricted geographic data from the Survey of Income and Program Participation (SIPP). The analysis employs a novel household-level instrument that measures the duration of homeowners' exposure to housing market booms driven by credit expansion in housing supply-constrained areas, leveraging cross-household variation in both the timing and location (counties) of home purchases. Housing wealth negatively affects women's labor supply, a 1% increase lowers participation by 0.098 pp, but shows no significant effect for men. This negative wealth effect among female workers is driven primarily by childcare responsibilities and human capital investment, as it is strongest among mothers of young children and those who report child-related reasons for not working. Other potential mechanisms, such as income effects, precautionary saving, or liquidity constraints, do not seem to fully explain the negative association.

Keyword: Housing Wealth, Credit Expansion, Female Labor Supply, Childcare and Human Capital Investment

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1 Introduction

Housing wealth constitutes the largest component of household wealth for most families, with dramatic price fluctuations over the past fifteen years generating substantial wealth shocks for homeowners (Campbell, 2006; Pfeffer and Waitkus, 2021). This makes housing a uniquely powerful channel through which economic fluctuations influence individual work decisions. This paper examines the causal effect of housing wealth on labor supply decisions, a relationship with important implications for both estimating fundamental behavioral parameters and understanding aggregate economic outcomes. Understanding these effects has broader implications for economic modeling, as housing booms and busts can influence aggregate labor market activity, with prior research documenting how housing cycles affect local employment outcomes (Mian et al., 2013; Charles et al., 2018). These wealth shocks also provide critical insight into income effects via the Slutsky equation, which is essential for evaluating taxation and welfare programs that affect work incentives (Saez, 2001; Moffitt, 2002; Keane, 2011b). Despite its importance, consensus on the magnitude and mechanisms of wealth effects on labor supply remains limited due to challenges in isolating exogenous wealth variation.

This paper addresses this identification challenge by providing credible estimates of how individuals adjust their labor supply in response to housing wealth changes. Using confidential 1990–2004 Survey of Income and Program Participation (SIPP) data with restricted county identifiers, I link detailed household information to local market conditions. The data enable the construction of a novel household-level instrument measuring homeowners' exposure duration to housing booms driven by exogenous credit expansion in housing supply-constrained areas, leveraging variation in both purchase timing and location (counties). This dual-source variation exploits differences in housing wealth accumulation that are plausibly exogenous to household characteristics or local economic conditions, enabling comparison of workers with identical characteristics but different home equity growth due solely to location and home purchase timing.

My results reveal a significant negative relationship between housing wealth and labor supply, with effects concentrated among women. Specifically, a 1% increase in housing wealth reduces women's labor force participation by 0.098 percentage points, while effects for men are insignificant. The wealth effect is most pronounced among college-educated mothers with young children, suggesting that childcare responsibilities serve as a factor through which housing wealth affects labor supply decisions. These gender-differentiated responses complement work by Goldin (2014), Blau and Kahn (2017), and Kleven et al. (2019) on factors driving persistent gender gaps in the labor market, and suggest that housing market fluctuations may influence gender gaps in labor force participation. These findings enhance our understanding of how housing market dynamics transmit to labor markets and provide valuable insights for policymakers navigating volatile housing markets and their broader economic consequences.

Theoretically, housing wealth affects labor market decisions through several channels that vary in their gender specificity. Three gender-neutral mechanisms provide the foundation for understanding

aggregate effects. First, increases in non-labor wealth raise consumption of normal goods, including leisure, thereby reducing labor supply as individuals substitute leisure for work (Campbell and Cocco, 2007; Cesarini et al., 2016, 2017). Second, housing wealth relaxes borrowing constraints through financial innovations like home equity loans, enabling households to finance consumption without requiring additional work income (Mian and Sufi, 2011; Cooper, 2013; DeFusco, 2015; Aladangady, 2017). Third, housing wealth can serve as precautionary saving, reducing the need for liquid buffers against income risk and thereby increasing consumption in environments with incomplete insurance markets (Painter et al., 2022). While these income, liquidity, and precautionary channels predict a general negative relationship between housing wealth and labor supply for both men and women, they cannot fully explain the pronounced gender and education heterogeneity observed empirically—particularly the strong responses among college-educated women with young children.

Two complementary, gender-specific mechanisms help explain these differential effects. First, the child human capital investment mechanism reveals a striking pattern that contradicts standard opportunity cost predictions: college-educated mothers exhibit the strongest labor supply reductions following housing wealth gains when childcare demands are high, despite facing the highest forgone earnings from leaving the workforce (Blundell and MaCurdy, 1999a; Anderson and Levine, 2000; Gelbach, 2002). This pattern reflects strong preferences for time-intensive child human capital investments rather than opportunity cost considerations. A substantial literature documents that highly educated parents invest significantly more time and resources in children's development. College-educated mothers spend considerably more time on childcare than less-educated mothers (Guryan et al., 2008), engage in intensive parenting practices and "concerted cultivation" strategies (Ramey and Ramey, 2010; Lareau, 2003), and make greater investments in early childhood education and enrichment activities (Doepke and Zilibotti, 2019; Kalil et al., 2012). Wealth enables these mothers to act on these investment preferences by prioritizing direct parental care over market work when financial pressures are alleviated. Critically, this effect is most pronounced when both parents are college-educated, suggesting that shared educational backgrounds and child investment preferences within households amplify the response (Doepke and Zilibotti, 2019). This mechanism operates primarily for women, who bear disproportionate childcare responsibilities even in highly educated dual-career households. Second, housing wealth functions as a commitment device through "collateralized marriage," providing insurance for lower-earning partners—typically women—against post-divorce financial risks (Lafortune and Low, 2017, 2023). This insurance effect amplifies household gender specialization patterns, with college-educated mothers in dual-college households leveraging their financial security to reduce market work while men show muted responses regardless of housing wealth changes (Lundberg and Pollak, 1994; Bertrand et al., 2015; Tertilt et al., 2016).

This paper contributes to the literature by recovering heterogeneity in labor supply responses to housing wealth, particularly emphasizing the opportunity cost/childcare and commitment device mechanisms, which drive significant effects among college-educated mothers with young children but not non-college workers. By integrating family economics with traditional wealth effect models, it eluci-

dates the mechanisms behind these differential responses, highlighting their empirical importance.[1]

Existing evidence on housing wealth effects remains limited, and estimating the causal effect of housing wealth on labor supply faces significant endogeneity challenges that complicate empirical identification. At the individual level, omitted variable bias arises because unobserved household characteristics, such as ability or financial literacy, can affect labor supply decisions and housing wealth conditions simultaneously. For example, individuals with higher ability may be more likely to participate in the labor force and make wise financial decisions, leading to greater housing wealth accumulation. At the local level, economic conditions, demographic changes, and other regional factors simultaneously influence both house prices and labor market outcomes, making it difficult to isolate the independent effect of wealth changes. Positive economic shocks increase both labor demand and housing values, generating spurious positive correlation, while negative shocks could trigger added-worker effects among secondary earners even as housing values decline. Reverse causality presents an additional challenge, as households with better labor market outcomes are more likely to accumulate housing wealth through higher incomes and access to better mortgage terms, while those anticipating labor market exit may strategically downsize their housing. Finally, residential sorting complicates identification because households may choose locations based on both housing market prospects and employment opportunities, creating spurious correlations between housing wealth and labor outcomes. This interplay between unobserved characteristics, labor market outcomes, and housing wealth can significantly bias OLS estimates, necessitating careful empirical strategies that can credibly isolate exogenous variation in housing wealth to identify causal effects on labor supply.

These endogeneity challenges explain the mixed results in the literature and underscore the need for sophisticated identification strategies. To address these issues, researchers have employed various methodological approaches. Instrumental variables (IV) strategies are particularly prevalent, leveraging plausibly exogenous variation in house prices or local housing supply elasticity as instruments for household housing wealth (Henley, 2004; Disney and Gathergood, 2018; Dettling and Kearney, 2014; Chetty et al., 2017; Aladangady, 2017; Farnham, 2007; Zhao and Li, 2017; Begley and Chan, 2018; Asquith, 2025). However, these instruments face significant limitations. Whether based on housing prices or supply elasticity, they often fail to adequately account for endogenous residential sorting patterns. High-ability households may systematically select into areas with limited land availability, such as coastal regions, thereby confounding estimates of labor market effects (Ferreira and Gyourko, 2012). This sorting mechanism creates additional bias that is difficult to separate from the housing wealth effects of interest. Alternative approaches have exploited natural experiments, such as policy changes that affect house prices while remaining plausibly unrelated to labor market conditions (Li et al., 2020). While these studies offer valuable insights, they are often context-specific and may not generalize broadly. Despite these methodological advances, obtaining credible estimates of hous-

[1]Housing wealth variations can also affect homeowners' location and geographic migration decisions, influencing job opportunities Bennett et al. (2001); Lovenheim (2011). To focus on wealth effects, this study only includes households that have not moved during the sample period. Consequently, mobility is not considered in this analysis but will be explored in future research.

ing wealth effects remains challenging. The difficulties are particularly pronounced when examining heterogeneous effects across demographic groups, where selection issues—including endogenous residential sorting—may be especially critical and harder to control for through standard econometric techniques.

This paper introduces a novel household-level instrumental variable measuring homeowners' exposure duration to housing market booms driven by credit expansion in supply-constrained areas. The instrument combines three components: (1) state-level bank deregulation status (local demand shocks), (2) county-level proportion of developable land (local supply elasticity), and (3) household-level home purchase timing (temporal variation). These components jointly identify how long each homeowner experienced local housing booms generated by the interaction of credit expansion and inelastic housing supply.

The first two components capture important demand and supply factors that contribute to local housing market booms and geographic variations in the 1990s and 2000s. The demand component leverages credit expansion effects on housing market demand and price booms, while the supply component exploits how supply elasticity—shaped by land use regulations and geographical constraints—mediates price responses to demand shocks.[2] Unlike prior studies using MSA- or state-level housing supply approximations, this study employs refined county-level land availability data for greater granularity. The geographic variation captures differences in local housing markets across counties, identifying how similar households in different locations experience divergent housing wealth trajectories.

The third component uses household-level home purchase timing from the restricted geo-coded Survey of Income and Program Participation (SIPP), merging it with state-level bank deregulation and county-level land availability data. This household-level temporal variation provides key identification advantages. First, it addresses endogenous residential sorting concerns by comparing households within the same geographic area who differ only in purchase timing, where timing is largely determined by life-cycle factors rather than strategic decisions to maximize housing wealth appreciation. This approach controls for time-invariant local characteristics while creating quasi-random variation in exposure duration, where two equally capable households in the same county experience different housing wealth trajectories based on different purchase years. Second, by capturing variation at the individual household level, it enables more precise identification of causal effects and analysis of heterogeneous responses across demographic characteristics.

This dual-source variation leverages housing wealth differences that are plausibly exogenous to household characteristics or local economic conditions, enabling comparison of workers with identical observable characteristics but different housing wealth due to location and purchase timing. By isolating housing wealth effects from other economic influences and controlling for location-specific

[2]See Favara and Imbs (2015); Chu (2017); Maggio and Kermani (2017); Saadi (2020); Mian and Sufi (2021); Choi and Hansz (2021); Yang (2024a,b) for evidence on how credit expansion influences housing market demand and drives price booms. For further details on how supply elasticity, shaped by land use regulations and geographic constraints, mediates price responses to demand shocks, see Quigley and Raphael (2005); Glaeser et al. (2008); Gyourko et al. (2008); Saiz (2010); Gyourko and Molloy (2015); Song (2021); Gyourko et al. (2021); Ferreira and Gyourko (2023).

factors, this methodology mitigates biases from endogenous residential sorting. This approach yields more precise causal estimates than previous local-level instruments, offering a robust tool to address selection issues and enhance the accuracy of housing wealth effect estimates.

This study leverages the confidential 1990-2004 panels of the Survey of Income and Program Participation (SIPP), which include restricted-use county identifiers, making it ideal for this analysis for three reasons. First, the SIPP provides a nationally representative sample with detailed data on labor market outcomes and household assets, including housing wealth. This enables a robust analysis of the impact of housing wealth on outcomes such as labor force participation and hours worked.[3] Second, restricted-use county identifiers and home purchase timing data allow for precise integration with state-level bank deregulation and county-level land availability.[4] This supports the construction of a household-level instrumental variable (IV) that captures exposure to housing booms driven by credit expansion (demand) and land use restrictions (supply). Third, the SIPP's rich demographic and financial data facilitate tests of alternative mechanisms via non-housing wealth, financial and borrowing channels. These variables enable subsample analyses to identify responsive demographic groups, such as those defined by gender, education, presence of young children, or borrowing constraints, clarifying the mechanisms through which housing wealth shapes labor market behavior across different populations.

This study uncovers a significant negative relationship between housing wealth and labor supply, with pronounced effects for women. A 1% increase in housing equity is associated with a 0.069 percentage point increase in the probability of not participating in the labor force for the full sample, with a stronger effect for females (0.098 percentage points) and a positive but statistically insignificant effect for males (0.040 percentage points). Effects on other labor market outcomes, such as hours worked, part-time employment, or unemployment, are statistically insignificant. These gender differences remain robust across alternative measures of housing wealth, instrumental variable (IV) specifications, and additional controls for individual or household characteristics.

To examine the mechanisms driving the housing wealth effect on labor supply, particularly its pronounced gender differences, I conduct a series of empirical tests that provide direct evidence for the child human capital investment mechanism as the central channel. First, the estimated effects are statistically significant only among women with young children, consistent with the intensive childcare demands of early parenthood, and are negligible for men and parents of older children. Second, I leverage a unique SIPP question on self-reported reasons for labor force non-participation. Instrumental-variable estimates reveal that higher housing wealth significantly raises the probability of non-participation, specifically for child-related reasons, while showing no significant association with non-child-related reasons such as health, education, retirement, or caregiving for elderly or disabled family members. These complementary findings firmly establish the child human capital investment mechanism as the primary channel through which housing wealth affects women's labor supply.

[3]The effect of housing wealth on retirement is discussed in detail in Yang (2025).

[4]Other data sources and measures of local housing supply elasticity are explored in robustness checks.

To further support the child human capital investment mechanism and evaluate alternative explanations, I conduct heterogeneity analyses across women's education level, spouse's education level, marital status, and household financial conditions (including non-housing liquid wealth). The results are strongest among college-educated mothers and in households where both parents hold college degrees—patterns inconsistent with a standard income effect (given college-educated women's typically lower labor supply income elasticity) or precautionary saving motives (as dual-college households generally enjoy higher, more stable incomes). Instead, these findings align with the greater valuation of intensive childcare and child human capital investment among highly educated parents, leading women to reduce labor supply more readily when housing wealth increases. The effect is also more pronounced among married women than never-married ones, consistent with commitment or intra-household bargaining effects that facilitate household specialization, and somewhat larger in households with low non-housing wealth, suggesting liquidity constraints play a secondary amplifying role. To further strengthen the causal interpretation, I test the instrument's impact on non-housing wealth and household borrowing; both outcomes are statistically insignificant, supporting the exclusion restriction and indicating that relaxed borrowing constraints are not the primary driver of the observed labor supply responses. Overall, the decline in female labor supply is concentrated among college-educated married women with young children, highlighting the interaction of intensive childcare demands, child human capital investment preferences, and housing wealth's role as a commitment device enabling intra-household specialization (Lafortune and Low, 2017, 2023; Sun et al., 2023). These results underscore how housing wealth interacts with gender roles and family dynamics to shape labor market behavior.

The paper proceeds as follows: Section 2 reviews the related literature, while Section 3 discusses potential mechanisms. Section 4 examines U.S. housing price dynamics driven by bank branching deregulation and supply elasticity, presents a conceptual framework linking housing wealth to labor supply, and provides motivating empirical evidence. Section 5 describes the data and sample statistics. Section 6 details the instrumental variable strategy, and Section 7 presents the main results, robustness checks, and heterogeneous effects. Section 8 explores underlying mechanisms, and Section 9 concludes.

2 Literature Review

2.1 Non-Housing Wealth or Unearned Income and Labor Market Outcomes

The literature on non-housing wealth and unearned income effects on labor market outcomes utilizes exogenous shocks, such as lottery winnings and inheritances, to identify causal impacts, consistently finding that wealth increases reduce labor supply, though effect sizes differ by context. Studies on lottery winnings, including Imbens et al. (2001), Cesarini et al. (2017), Picchio et al. (2018), and Golosov et al. (2024), report negative wealth effects on earnings and labor participation, with U.S. responses notably stronger than those in Sweden and the Netherlands. Similarly, research on inheritances and bequests,

such as Holtz-Eakin et al. (1993), Joulfaian and Wilhelm (1994), Brown et al. (2010), Holtz-Eakin et al. (1994), and Andersen and Nielsen (2012), documents reduced labor supply following wealth shocks. Natural experiments involving transfer income, as explored in Krueger and Pischke (1992), Bengtsson (2012), Jacob and Ludwig (2012), Gelber et al. (2016), Feinberg and Kuehn (2018), and Jones and Marinescu (2019), further support these findings. Regarding gender effects, the literature reveals contradictory findings: Feinberg and Kuehn (2018) and Imbens et al. (2001) find stronger labor supply reductions for women, particularly married women, and Andersen and Nielsen (2012) notes larger effects for single women without children; however, recent studies like Cesarini et al. (2017) and Golosov et al. (2024) report no significant gender differences, highlighting the variability in gender-specific responses across contexts.[5]

This paper contributes to the literature by examining labor supply responses to housing wealth, which differs fundamentally from lottery winnings and inheritances. Unlike these relatively rare windfalls that provide immediate liquidity, housing wealth represents the largest component of household wealth for most American families (Campbell, 2006). Consequently, housing wealth shocks may have broader and more substantial effects on labor supply across the general population. Moreover, the illiquidity of housing wealth introduces distinct dynamics. Although lottery and inheritance benefits often lead to rapid reductions in labor supply due to their lump sum nature (Imbens et al., 2001; Cesarini et al., 2016), the effects of housing wealth depend on the ability of households to access equity or their perceptions of lifetime wealth (Cooper, 2013). This illiquidity can result in smaller or delayed labor supply adjustments compared to financial windfalls. Furthermore, housing wealth is subject to market-driven fluctuations, generating idiosyncratic and exogenous shocks that differ from the more predictable nature of inheritances (Imbens et al., 2001). Understanding these distinctions is crucial for explaining differential responses to labor supply between wealth sources and for understanding gender heterogeneity effects.

2.2 Housing Wealth effects on Labor Market

This paper contributes to the literature by addressing the poorly understood effect of housing wealth on labor supply, a relationship of significant economic importance for which reliable U.S. evidence remains limited. Most studies document negative wealth effects, consistent with leisure being a normal good.

[5]Another strand of literature on labor supply responses to unearned income employs structural models to estimate income effects and elasticities from observational data on variations in unearned income, wages, and tax rates. As reviewed in Pencavel (1986), Killingsworth and Heckman (1986), Blundell and MaCurdy (1999b), Keane (2011a), and Saez et al. (2012), these studies often use spousal or capital income changes, or tax reforms, as instruments for unearned income shifts. However, challenges in ensuring exogeneity and separating income from substitution effects without strong assumptions on preferences, functional forms, and unobservables lead to varied estimates and no consensus on income effect magnitudes. Regarding gender effects, Pencavel (1986) notes gender-specific factors in men's labor supply, Killingsworth and Heckman (1986) highlights fertility impacts on female labor supply, Blundell and MaCurdy (1999b) and Keane (2011a) address gender variations (with Keane (2011a) also noting fertility effects), while Saez et al. (2012) does not explore gender distinctions.

U.S. evidence has focused primarily on retirement decisions rather than labor supply among working-age workers. Research on retirement decisions or older households yields mixed results: Farnham (2007) finds negative housing wealth effects on labor supply among older households, with stronger effects for men; Zhao and Burge (2017) and Zhao and Li (2017) report positive effects on retirement probability for women but not men; Begley and Chan (2018) finds no significant gender differences but weak negative effects on labor force participation; and Asquith (2025) identifies negative effects, particularly for older women nearing retirement. Using British data, Henley (2004) finds that housing wealth appreciation reduces working hours for both men and women. Similarly, Disney and Gathergood (2018) reports that in Britain, housing wealth gains significantly decrease labor supply, particularly among young married female homeowners. Meanwhile, several studies using data from China present a generally consistent picture of negative labor supply responses to housing wealth. Using different identification strategies, both Fu et al. (2016) and Li et al. (2020) find that rising housing wealth reduces labor force participation, with effects concentrated among females and younger individuals. Jiang et al. (2023) finds that relative housing wealth leads to a reduction in working hours, though it reports no significant gender differences.[6]

The lack of evidence in the U.S. context, combined with mixed findings on gender effects, underscores the need for robust identification strategies and further research. This paper advances the literature in three key ways: first, it fills the evidence gap by providing new insights into housing wealth effects among working-age workers within the U.S. context; second, it introduces a novel instrumental variable to more accurately capture household-level housing wealth variation; third, it provides a detailed analysis of subsamples and demographic groups, offering critical insights into the mechanisms driving these effects.

2.3 Housing Wealth and Other Outcomes

A substantial literature examines how housing wealth affects household behavior, extending beyond labor supply to include consumption, household finances, and significant life decisions such as fertility and marriage. Studies show that house price fluctuations drive consumption changes through multiple channels: a pure wealth effect (homeowners spend more as they feel wealthier; Campbell and Cocco (2007)), a collateral effect (higher prices increase borrowing capacity; Aladangady (2017)), and precautionary saving (Painter et al., 2022). Housing wealth further affects financial behavior, reducing stock market participation (Chetty et al., 2017) and altering demand for insurance products (Davidoff, 2010), while also driving mortgage refinancing and credit demand (Cocco, 2005; Cooper, 2013; DeFusco, 2015). Critically, these liquidity and wealth effects spillover into demographic choices, easing credit constraints for education (Lovenheim, 2011; Lovenheim and Reynolds, 2013), raising

[6] Johnson (2014) suggests that U.S. housing price increases, signaling higher living costs, can increase labor supply, with no significant gender differences noted. This study is distinct as it focuses on housing price effects rather than housing wealth effects.

fertility rates (Lovenheim and Mumford, 2013; Dettling and Kearney, 2014), and stabilizing marriages (Farnham et al., 2011; Klein, 2017)—all factors indirectly tied to labor supply decisions. At the macroeconomic level, housing wealth amplifies business cycle dynamics, with leveraged households exhibiting sharper consumption declines during downturns (Mian and Sufi, 2011), a pattern formalized in structural models (Iacoviello, 2011).

Addressing endogeneity remains a persistent challenge in identifying housing wealth effects on labor supply and other outcomes. These methodological difficulties contribute to the mixed findings in the literature and underscore the need for robust identification strategies. Researchers commonly employ instrumental variables (IV) approaches that leverage plausibly exogenous variation in house prices or local housing supply elasticity. Early studies relied on local housing price indices as instruments (Henley, 2004; Disney and Gathergood, 2018; Farnham, 2007; Zhao and Li, 2017; Begley and Chan, 2018), but housing prices themselves can be endogenous and correlated with unobserved local economic conditions, likely biasing causal effect estimates. For example, Ferreira and Gyourko (2012) demonstrated that socioeconomic factors correlate with varying housing boom durations and magnitudes at the local level. More recent studies have utilized housing supply elasticity estimates from Saiz (2010) and Gyourko et al. (2008) as instruments for price changes during the 2000s housing boom and bust, making this approach increasingly popular (Dettling and Kearney, 2014; Chetty et al., 2017; Aladangady, 2017; Asquith, 2025). These instruments, based on geographic and regulatory constraints, face their own challenges, as lower elasticity cities may exhibit greater cyclicality due to industrial composition (Davidoff, 2016). Moreover, their explanatory power diminishes for periods before 2000 (Guren et al., 2021).

Moreover, relying solely on cross-regional variation as an instrumental variable for household housing wealth—whether based on housing prices or supply elasticity—may introduce bias from other unobserved regional factors that correlate with both housing and labor markets. Endogenous residential sorting presents an additional source of potential bias, as high-ability households may be more likely to locate in deregulated areas with limited land availability, such as coastal regions, thereby confounding estimates of labor market effects. In response to these limitations, alternative approaches have emerged. Recent innovations include sensitivity instruments developed by Guren et al. (2021), which estimate local house price sensitivity to regional cycles. Natural experiments, such as those exploiting housing policy discontinuities in China (Li et al., 2020), yield valuable context-specific insights, yet their findings may not readily apply to other settings.

While the existing literature highlights the broad economic significance of housing wealth, this study advances the field by introducing a novel instrumental variable that varies at both the household and local levels to better identify causal effects on labor supply decisions. The instrumental variable strategy thus leverages exogenous variation across locations, years, and purchase timing to provide robust identification of housing wealth effects, with the household-level timing component serving as a powerful tool to isolate housing wealth variation from confounding factors while enabling detailed analysis of heterogeneous treatment effects. This methodological innovation could potentially be

extended to analyze other housing wealth effects on household behavior, such as consumption, savings, and major life decisions.

3 Potential Mechanisms

The theoretical foundation for analyzing housing wealth's impact on labor supply draws from the life-cycle and permanent income hypotheses, which predict that wealth changes should influence consumption and labor decisions over time. However, the relationship between housing wealth and labor supply operates through multiple, often interconnected mechanisms that can be categorized into **gender-neutral** and **gender-specific mechanisms** that help explain both aggregate effects and the substantial heterogeneity observed across demographic groups. First, three gender-neutral mechanisms provide the theoretical foundation for why housing wealth should affect labor supply generally.

Income Effect: Basic economic theory suggests that increases in non-labor income or wealth raise the consumption of normal goods (Imbens et al., 2001; Blundell and MaCurdy, 1999a). Leisure, like consumption, is typically considered a normal good. Therefore, housing wealth gains should lead to increased leisure and decreased labor supply for homeowners of both genders, while housing wealth losses should have the opposite effect. Studies based on microdata often find that the aggregate marginal propensity to consume out of housing wealth is small (Campbell and Cocco, 2007; Case et al., 2005), which could reflect that housing wealth changes primarily affect labor supply decisions rather than consumption patterns, particularly among older workers who may use housing gains to finance earlier retirement (Zhao and Li, 2017).

Relaxing of Liquidity and Borrowing Constraints: Housing wealth serves as a critical form of household collateral that relaxes borrowing constraints through two complementary channels that operate equally for both genders. First, the collateral value of housing enables access to secured credit even for liquidity-constrained households. Second, financial innovations since the 1990s—including the proliferation of home equity loans, lines of credit (HELOCs), and cash-out refinancing—have dramatically enhanced the liquidity of housing wealth (Mian and Sufi, 2011; Cooper, 2013; DeFusco, 2015; Aladangady, 2017). These developments allowed homeowners to more readily convert home equity into consumable resources, with extracted equity rising from 2% of U.S. personal income in 1990 to over 11% by 2004 (Greenspan and Kennedy, 2008).

Precautionary Saving: Housing wealth can function as precautionary saving, reducing the need for liquid buffers against income risk (Painter et al., 2022). When housing wealth increases, households may feel more financially secure and reduce their precautionary labor supply, consuming more or working less to smooth utility over time in environments with incomplete insurance markets.

While the three gender-neutral mechanisms account for the overall negative relationship between housing wealth and labor supply, they cannot adequately explain the strong heterogeneity observed in the data. Empirically, the effect is almost entirely concentrated among women, particularly college-

educated mothers with young children, whereas male labor supply shows little to no response to changes in housing wealth. Two complementary gender-specific mechanisms help explain the pronounced demographic heterogeneity, particularly the concentration of effects among college-educated women with young children.

Childcare Responsibility and Human Capital Investment Mechanism: This mechanism reveals a striking pattern that contradicts standard opportunity cost predictions. College-educated mothers exhibit the strongest labor supply reductions following housing wealth gains when childcare demands are high, despite facing the highest forgone earnings from leaving the workforce (Blundell and MaCurdy, 1999a; Anderson and Levine, 2000; Gelbach, 2002). Rather than reflecting opportunity cost considerations—which would predict smaller responses among high-wage women—this pattern reflects strong preferences for time-intensive child human capital investments. A substantial body of evidence documents that highly educated parents invest significantly more time and resources in children's development. Guryan et al. (2008) show college-educated mothers spend considerably more time on childcare than less-educated mothers, with gaps widening over recent decades. Ramey and Ramey (2010) documents the rise of intensive parenting practices concentrated among educated families, while Lareau (2003) demonstrates that educated parents engage in "concerted cultivation"—actively fostering children's talents through organized activities and sustained reasoning. Doepke and Zilibotti (2019) and **?** further show that parental education strongly predicts investments in early childhood education, enrichment activities, and educational resources. These investment patterns reflect beliefs that parental time inputs yield high returns for child cognitive and behavioral outcomes. Housing wealth enables educated mothers to act on these investment preferences by reducing market work and prioritizing direct parental care when financial pressures are alleviated. Studies show that high childcare costs and quality concerns lead educated mothers to opt for home-based care when wealth permits, despite the availability of market alternatives (Baker et al., 2008; Blau, 2006; Morrissey, 2017). Critically, the effect is most pronounced when both parents are college-educated, suggesting that shared educational backgrounds and child investment preferences within households amplify the response (Doepke and Zilibotti, 2019). In contrast, non-college mothers in lower-wage roles show weaker responses, as ongoing liquidity needs or reliance on informal care arrangements constrain their adjustments, even though their lower opportunity costs should theoretically facilitate reductions (Edwards and Fogarty, 2005; Anderson and Levine, 2000). Effects are strongest with young children requiring intensive supervision but muted for older, more independent children (Baker et al., 2008; Gelbach, 2002). This mechanism is inherently gendered because women bear disproportionate childcare responsibilities even in highly educated dual-career households, making their labor supply decisions more sensitive to factors affecting the trade-off between market work and child-rearing time.

Housing Wealth as Commitment Device: While the income and liquidity mechanisms help explain the overall negative relationship between housing wealth and labor supply, they provide limited insight into the pronounced gender differences. Standard wealth effects and liquidity constraints should theoretically affect male and female labor supply similarly, yet our empirical results show

effects concentrated almost exclusively among women, particularly those with young children. This gender-specific pattern points to a mechanism involving intrahousehold dynamics, complementing the opportunity cost and childcare mechanism in explaining gender and college heterogeneity. This channel emphasizes that housing wealth is a commitment device within a couple's decision-making, affecting household specialization decisions—an aspect underexplored in labor supply literature. Lafortune and Low (2017, 2023) propose the concept of "collateralized marriage," where joint assets like housing enhance commitment by providing insurance to lower-earning partners, typically women, against adverse post-divorce outcomes. This commitment mechanism interacts with household bargaining dynamics, enabling greater specialization with women focusing more on non-market activities (Lundberg and Pollak, 1994; Bertrand et al., 2015; Tertilt et al., 2016; Sun et al., 2023). When housing wealth increases, it strengthens this commitment value, potentially amplifying existing patterns of gender specialization. The effect is strongest among college-educated mothers with young children, who have both the bargaining power to negotiate household specialization and strong preferences for child investment, allowing them to reduce market work when financially secure (Guryan et al., 2008; Ramey and Ramey, 2010). In contrast, non-college mothers may have less bargaining power, and men typically have weaker preferences for reducing market work regardless of housing wealth changes. Together with the opportunity cost and childcare mechanism, this perspective accounts for the observed gender and college asymmetries by linking housing wealth to household bargaining dynamics that incentivize childcare provision, particularly among educated women.

These mechanisms are not mutually exclusive and likely operate simultaneously, with their relative importance varying across households. The three gender-neutral mechanisms (income, liquidity, and precautionary saving) provide the foundation for understanding why housing wealth should affect labor supply generally and establish the baseline expectation of negative effects. However, the two gender-specific mechanisms—child human capital investment and commitment device—are crucial for explaining the pronounced demographic heterogeneity observed empirically. The concentration of effects among college-educated mothers with young children, particularly in dual-college households, suggests that these latter two mechanisms and their interaction may be empirically most important for understanding when and why housing wealth significantly affects labor decisions. The gender-neutral mechanisms predict symmetric effects that are clearly contradicted by the data, while the gender-specific mechanisms naturally generate the observed patterns of heterogeneity. This integration of family economics with traditional wealth effect models represents a key contribution to understanding labor supply heterogeneity.

4 Motivation Evidence: Sources of Exogenous Variation for IV

Housing is a critical component of U.S. household wealth, representing 30% of total household assets[7]. For middle-class households (those in the 20th to 90th wealth percentiles), it accounts for 50-60% of net worth, based on the 2022 Survey of Consumer Finances (SCF). The 2000s featured dramatic geographic variation in home price appreciation. Coastal markets like California (15% annual growth), Florida, and Nevada saw rapid gains, while interior markets like Dallas and Atlanta remained stable. This spatial heterogeneity provides a source of exogenous variation that helps identify the causal effect of housing wealth on labor market outcomes.

Both demand and supply factors played a role in the housing market boom and its geographic variation. On the demand side, exogenous mortgage credit expansion significantly increased house prices. Following U.S. bank branching deregulation (1994–2005), treated banks expanded credit through geographic diversification, which increased housing demand and raised prices. Studies by Favara and Imbs (2015) and Yang (2024b) use this deregulation as an instrument, establishing that these credit increases occurred for deregulated banks but not in placebo samples. This rules out demand-based explanations and cleanly identifies an exogenous credit supply shock.[8]

On the supply side, housing supply elasticity mediated the price response to these demand shocks. In areas with limited land availability and strict regulations, construction is constrained, causing demand shocks to translate into higher prices. In contrast, abundant land and fewer constraints allow new supply to stabilize prices (Quigley and Raphael, 2005; Gyourko et al., 2008; Glaeser et al., 2008; Saiz, 2010; Gyourko and Molloy, 2015; Song, 2021; Gyourko et al., 2021). While local geography and regulations do not explain all cross-sectional variation, they provide exogenous variation in how prices respond to a given demand shock.

This analysis leverages the interaction of two forces—exogenous credit supply shocks and exogenous differences in housing supply elasticity—which creates cross-regional variation that accurately captures the spatial pattern of housing price dynamics. However, using only cross-regional variation as an instrumental variable for household housing wealth may introduce bias from other unobserved regional factors that correlate with both housing and labor markets. Endogenous residential sorting also presents potential bias, as high-ability households may be more likely to locate in deregulated states with limited land availability, such as coastal areas, confounding estimates of labor market effects.

To mitigate these concerns, the analysis combines cross-county housing market variation with household-level timing of house purchases. This household-level temporal variation provides several key identification advantages. First, it addresses residential sorting and unobserved local heterogeneity by comparing households within the same geographic area who differ only in their purchase timing—a factor driven primarily by life-cycle events rather than strategic decisions. Second, it creates quasi-

[7]$56.0T of $186.8T in Q2 2025, according to Federal Reserve data

[8]More details on the bank branching deregulation are described in the Appendix. Other studies estimating credit expansion effects on housing prices include (Chu, 2017; Maggio and Kermani, 2017; Saadi, 2020; Mian and Sufi, 2021; Choi and Hansz, 2021; Yang, 2024a).

random variation in exposure duration to housing booms, where the specific year of purchase within a location generates exogenous differences in housing wealth accumulation unrelated to household ability or labor market preferences. Third, by capturing variation at the individual household level rather than aggregate geographic units, it allows for more precise identification of causal effects and enables analysis of household-level heterogeneity in responses—such as differential effects across education levels, family composition, or other demographic characteristics that vary within locations.

The instrumental variable strategy thus leverages exogenous variation across locations, years, and purchase timing to provide robust identification of housing wealth effects, with the household-level timing component serving as a powerful tool to isolate housing wealth variation from confounding factors while enabling detailed analysis of heterogeneous treatment effects.

Using Survey of Income and Program Participation (SIPP) data, Figure 1 shows that household housing equity began diverging after 1996, when major banking deregulation began. This geographic divergence persisted until 2005. Panel B shows that housing wealth growth was concentrated in deregulated areas with inelastic housing supply, as measured by developable land proportion. These results demonstrate the strong association between rising housing wealth and homeowner exposure duration to housing booms caused by bank deregulation in supply-restricted areas, motivating the IV construction in Section 6.2. Furthermore, Figure A3 confirms that other financial resources—including non-housing wealth, housing and non-housing debt, and total income—did not exhibit similar geographic or temporal patterns. This absence of parallel trends in other household financial outcomes reinforces the instrumental variable's validity.

5 Data and Summary Statistics

5.1 Survey of Income and Program Participation (SIPP)

This paper uses a confidential version of the Survey of Income and Program Participation (SIPP) 1991–2004 panels, which includes restricted-use county identifiers. The SIPP is a nationally representative survey that tracks 15,000 to 46,000 households (and their members) over 2- to 4-year periods. The 1991–2004 panels span 1991–2005, fully covering the periods before and after bank deregulation. The cross-county and cross-year variation provides sufficient exogenous variation in the instruments to identify and predict changes in household housing wealth.[9]

The SIPP features a unique panel design in which respondents are interviewed every four months (referred to as a "wave") about the preceding four-month period. Each wave includes a core module and one or more topical modules. The core module, administered in every wave, collects detailed demographic, employment, income, and housing tenure data at the household and individual levels.

[9]While the 1991–2004 SIPP panels cover the period 1991–2007, housing wealth information is available only for selected years: 1991, 1993, 1996–1999, and 2001–2005.

Topical modules, which rotate across waves, supplement the core data with information on specific subjects, such as assets and liabilities, typically appearing every three waves, or roughly annually.

The key variables of interest in this study are housing wealth measures from the SIPP's assets and liabilities topical module. Respondents report (1) their property's current market value ("How much do you think it would sell for today?") and (2) the outstanding mortgage principal. These variables enable calculation of housing equity as self-reported home value minus mortgage debt, yielding a household-level measure of housing wealth.[10]

Because housing wealth in the SIPP is self-reported, one potential concern is that respondents may report perceived rather than actual market values. However, Figure A2 demonstrates that self-reported home values serve as a reliable measure of housing wealth. The figure compares real growth rates of self-reported home values and housing wealth with national home equity appreciation (derived from the FHFA home price index), with both series deflated by the national consumer price index. The measures exhibit clear co-movement, and SIPP housing wealth closely tracks the national index, validating the accuracy of self-reported values. While self-reported values show slightly greater volatility—with somewhat optimistic assessments during the boom period and more pessimistic expectations during the market collapse—this behavioral pattern actually strengthens the case for using self-reported measures. Since homeowner financial decisions are likely driven by perceived rather than objective housing wealth, self-reported values may better capture the mechanism through which housing wealth affects economic behavior.[11]

The SIPP provides detailed monthly individual-level data on labor supply, employment status, and labor income, making it particularly valuable for studying how housing wealth affects labor market outcomes. The primary outcome of interest is labor force participation (the extensive margin of labor supply), defined as a binary indicator equal to 1 if an individual is either employed or unemployed but actively seeking work, consistent with standard labor economics definitions.

To capture the full range of labor market responses, I also analyze intensive margin measures including monthly and weekly hours worked (for employed workers), hourly wages (either directly reported or calculated as monthly earnings divided by hours worked), unemployment status (indicating whether individuals reported any unemployment during the interview period), and household income (which reflects aggregate labor supply effects). This distinction between extensive and intensive margins is theoretically important because housing wealth shocks may affect the decision to participate in the labor market differently than they affect hours worked or wage levels. The analysis employs several data adjustments to ensure robustness: hours worked are top-coded at 80 hours per week to minimize the influence of outliers, and all monthly labor market measures are averaged at the

[10]While home value alone is frequently used as a proxy for housing wealth, it reflects housing consumption rather than accumulated wealth. For most households, mortgage liabilities substantially offset home value, particularly in the years immediately following purchase. Consequently, I use net home equity, the value after debt deduction, to more accurately capture housing wealth.

[11]The SIPP's wealth data quality is assessed in Czajka et al. (2003) and Eggleston and Klee (2015), who find it comparable to the Survey of Consumer Finances. More discussion on the wealth variable measurements and time trends in the SIPP can be found in Yang (2024a).

individual-year level to be consistent with the annual frequency of housing wealth data in the SIPP's survey design.

The analysis incorporates comprehensive control variables from multiple sources. The SIPP provides detailed demographic characteristics of individuals, including age, race, ethnicity, gender, and educational attainment. At the household level, controls include marital status, number of children, and non-housing financial variables including both assets and debts. Asset measures cover liquid holdings (cash, savings, and checking/deposit accounts) and semi-liquid assets (stocks, businesses, and non-residential properties), while debt measures include mortgage liabilities, consumer debt (credit cards and auto loans), and other outstanding obligations. These financial measures serve dual purposes as both control variables and mechanism channels, enabling clear differentiation between housing wealth effects and other household financial components.[12] All monetary values are adjusted for inflation using 1996 as the base year. To account for broader economic conditions, the analysis incorporates state-level unemployment rates along with employment and wage data from the Bureau of Labor Statistics' Quarterly Census of Employment and Wages (QCEW). This empirical strategy is designed to identify the causal effect of housing wealth on labor outcomes, controlling both individual-level characteristics and macroeconomic conditions.

The analysis sample drawn from the SIPP comprises individuals aged 15 to 55 observed between 1990 and 2007. Several sample restrictions are implemented here. Households that relocated during the sample period are excluded to avoid confounding mobility effects with housing wealth changes. The analysis further excludes renters and households with missing housing wealth data, as the study specifically examines homeowners' responses to variation in housing wealth. To minimize outlier effects, the sample excludes observations in the top and bottom 1 percent of the income distribution, along with extreme values in both financial wealth-to-income and housing equity-to-income ratios. The asset variables derive from topical modules available for 1991-2004 (excluding 1990, 2006, and 2007 due to data unavailability on the topical module that includes housing wealth). These selection criteria produce a final cross-sectional sample of approximately 146,000 observations, including 77,000 females and 69,000 males.[13]

5.2 Other Datasets

The paper has used other datasets that capture local housing and labor market conditions. For housing market conditions, I use the Federal Housing Finance Agency (FHFA) housing price indexes.[14] The FHFA indexes series are available from 1975 through 2015, constructed from all repeat-sale, single-family homes whose mortgages have been securitized by Fannie Mae or Freddie Mac each year. These data are available at the state and MSA levels and are widely used in the housing literature. I use

[12]While the SIPP primarily tracks income and program participation, its detailed wealth and debt data also model eligibility for means-tested programs.

[13]Sample sizes are rounded to comply with RDC disclosure requirements.

[14]These data can be accessed at http://www.fhfa.gov.

the county-level housing price index, which became available more recently, to better capture housing price variation at a more refined geographic unit.

To measure the geographic determinants of housing supply elasticity, I use the newly developed measures of the percentage of developable land in Lutz and Sand (2019). This measure expands on the popular topological land unavailability proxy from Saiz (2010) in three dimensions. First, it uses higher-resolution satellite imagery from the United States Geological Survey. Second, it provides more precise geographic boundaries. Third, this measure is available at multiple levels of disaggregation. Land unavailability is used to proxy the housing supply elasticity and test whether house prices increase more in less-elastic areas during the deregulation periods. Particularly, a state or a county is defined as having less land if the percentage of developable land is less than 70 percent of the total area. Specifically, land unavailability in Lutz and Sand (2019) measured at county levels shows no significant correlation with key economic indicators such as household income, business payrolls, employment, number of households, business establishments, college education share, foreign population share, and an amenities index, across different periods (2000/2001 to 2019). This lack of correlation between land unavailability and housing demand factors, especially at the county level, strengthens the case for using land unavailability as a component to construct instrument for household wealth.

As an alternative measure of housing supply constraints, I utilize metropolitan statistical area (MSA)-level indicators: the Wharton Residential Land Use Regulatory Index (WRLURI) from Gyourko et al. (2008) and the proportion of undevelopable land from Saiz (2010). The WRLURI, constructed from a 2005 survey of municipal officials, quantifies the stringency of local land use regulations. However, since WRLURI data are only available for approximately 200 MSAs, I employ these measures primarily for robustness checks rather than in my main analysis. The results remain qualitatively robust but show weaker statistical significance compared to my primary county-level measures. This attenuation may be due to the coarser geographic resolution of the MSA-level data relative to the county-level variation analyzed in Lutz and Sand (2019).

For local labor market conditions, I use the Quarterly Census of Employment and Wages (QCEW), which provides information on employment and wages reported by employers covering more than 95 percent of U.S. jobs. These variables are available at the county level and by industry.

5.3 Summary Statistics

Table 1 presents summary statistics from the Survey of Income and Program Participation (SIPP) for the years 1991-2004, offering a comprehensive overview of individual, household, and local labor market characteristics. The data is broken down into four columns: Total sample, Female subsample, Male subsample, and the difference between female and male subsamples.

Regarding Individual Variables, the average age of the sample is about 41 years, with males slightly older than females. The sample is roughly evenly split between genders (52% female). A high proportion (84%) of the sample is married, with no significant difference between genders.

On average, individuals have 1.2 children. The sample is predominantly white (88%), with small proportions of Black (7%) and other races (5%). Educational attainment is high, with 57% having a college education and 12% with graduate degrees. Notably, females have a higher college education rate but a lower graduate education rate compared to males. Labor market outcomes show significant gender differences. 18% of the total sample is not in the labor force, but this rises to 27% for females compared to only 8% for males. Males work more hours per week (44.07 vs. 37.11) and per month (191.3 vs. 161.1) than females. There's also a substantial gender wage gap, with males earning an average hourly wage of $28.48 compared to $21.31 for females.

Regarding household-level variables, the average household income is $92,540, with male-headed households having slightly higher incomes. The average house tenure is about 6.4 years, with female-headed households having slightly longer tenures. The average house purchase year is 1996 for both genders. House wealth averages $118,200, with no significant gender difference. However, male-headed households have higher house debt ($114,000 vs. $110,700) and non-house debt ($32,050 vs. $30,830). Non-house wealth averages $93,370, with a small advantage for male-headed households. About 83% of the sample lives in metro areas. The average employment in the local labor market (measured at the county) is 7,594, with no significant difference between genders. The average QCEW wage is $46,350, again with no significant gender difference. Statistical significance of gender differences is indicated by asterisks, with most differences being highly significant, suggesting robust gender disparities across various socioeconomic indicators.

6 Empirical Method

6.1 Ordinary Least Squares (OLS) Model

To examine housing wealth effects on labor market decisions, I start with the following OLS estimation model:

$$Y_{ict} = \beta_1 LogHW_{it} + \beta_2 X_{it} + \mu_c + \gamma_t + e_{ijt} \tag{1}$$

where Y_{ijt} is labor market outcomes for individual i living in location c in time period t, $LogHW_{it}$ is the log of the amount of housing equity. The main labor market outcome is labor force participation. X_{it} is a vector of control variables that includes individual demographic variables, such as age, race, and education, and household-level variables including marital status, number of children, household income, and other financial assets. The model also includes county and year-fixed effects as well as state time trends to control factors that simultaneously affect labor market outcomes and housing wealth. The coefficient of interest is β_1, which is interpreted as the responses of labor market choices about housing wealth. Standard errors are clustered at the household level to accommodate within-household correlation.

While the OLS estimates presented above document a significant association between housing wealth and labor market outcomes, they are unlikely to represent unbiased estimates of the causal relationship due to several fundamental identification challenges. The first concern stems from potential omitted variables operating at both individual and local levels that may correlate with both housing wealth changes and labor force participation decisions. At the individual level, unobserved heterogeneity in preferences, ability, or financial sophistication could simultaneously influence both housing investment decisions and labor supply choices. For instance, individuals with higher unobserved ability may be more likely to participate labor force while also being more likely to purchase homes in high-appreciation housing markets, thereby introducing upward bias into the OLS estimates.

More critically, unobserved local economic shocks present a systematic source of bias that could operate in either direction. Positive economic shocks to a region would simultaneously increase labor demand—thereby raising participation probabilities—and drive up local housing values, creating a spurious positive correlation between housing wealth gains and labor force participation. Conversely, negative economic shocks could generate bias through two mechanisms: first, by reducing labor demand and increasing involuntary exits from the labor force; and second, by triggering added-worker effects,[15] particularly among secondary earners (typically female workers), who enter the labor force to compensate for primary earners' job losses even as local housing values decline.

The second major identification challenge arises from reverse causality. Since housing wealth represents the market value of property net of outstanding mortgage debt, labor market outcomes may themselves be important determinants of housing wealth accumulation rather than merely responding to it. This reverse relationship operates through several channels. Labor force participants typically maintain higher and more stable incomes, enabling them to qualify for larger mortgages and purchase homes in more desirable neighborhoods with stronger appreciation potential. Additionally, individuals anticipating labor market exit may strategically downsize their housing, thereby realizing capital gains that appear in our data as housing wealth changes preceding labor force exit.

The endogenous relationship between labor market outcomes and housing wealth accumulation fundamentally complicates the interpretation of our OLS estimates and necessitates an identification strategy that can isolate exogenous variation in housing wealth. To address these endogeneity concerns and estimate the causal effects of housing wealth on labor market outcomes, this paper develops a novel instrumental variables (IV) approach. This strategy exploits variation in housing wealth that is plausibly exogenous to individual household characteristics and local economic conditions, thereby enabling more credible causal inference.

[15]The added-worker effect refers to the phenomenon where secondary earners (typically spouses) increase their labor supply in response to a primary earner's job loss or income shock. Early theoretical work by Lundberg (1985) established the microfoundations of this effect, showing that married women's labor supply responds significantly to husbands' unemployment. Subsequent studies (Stephens, 2002; Cullen and Gruber, 2000) have quantified these responses and shown how unemployment insurance attenuates the effect. Recent evidence (Halla et al., 2020; Juhn et al., 2007) suggests heterogeneity across households and potential weakening over time.

6.2 Instrumental Variable

I develop a novel instrumental variable approach that exploits the duration of homeowners' exposure to housing market booms driven by credit expansion in supply-constrained areas (exogenous demand and supply shocks in the housing market). The instrument combines three key components: (1) state-level bank deregulation status, capturing exogenous demand shocks from credit expansion; (2) county-level housing supply constraints, reflecting local supply inelasticity; and (3) household-level year of house purchase, providing temporal variation in exposure across households. Together, these components identify how long each homeowner experienced the local housing boom generated by the interaction of credit expansion and inelastic housing supply.

Specifically, the instrument Z_{ict} for household i in county c in year t is

$$Z_{ict} = \sum_{j=1990}^{j=t} E_{icj} = \mathbb{I}(j \geq \text{DeregYear}_c) \cdot \mathbb{I}(\text{Inelastic}_c = 1) \cdot \mathbb{I}(j \geq Purchase_i) \tag{2}$$

where:

- $E_{ic} = 1$ if household i live in county c where bank deregulation has occurred, county c has inelastic housing supply, and the household purchased their house before year t;
- $\mathbb{I}(\cdot)$ is the indicator function (1 if the condition holds, 0 otherwise);
- DeregYear_c is the year bank deregulation occurred in county c;
- Inelastic_c indicates housing supply elasticity status in county c;
- $Purchase_i$ is the year household i purchased their home.

To construct the instrument Z_{ict}, I make use of several key data sources. First, I obtain information on interstate bank branching deregulation by state and year from Rice and Strahan (2010). Second, I incorporate local housing supply elasticity measures based on the proportion of undevelopable land provided by (Lutz and Sand, 2019). The third component is the timing of home purchase, obtained from the SIPP.[16]

Crucially, I use the restricted geocoded version of the SIPP, which provides county-level FIPS codes.[17] These restricted geocodes are essential for precisely matching households to county-level variables critical for identifying housing wealth variation, as the literature documents that housing market dynamics vary substantially across counties due to differences in local-level housing supply

[16]This information is typically unavailable in standard household surveys such as the ACS or CPS. The SIPP is one of the few household-level datasets that records the year of home purchase for homeowners, making it essential for constructing the instrumental variable for housing wealth.

[17]These data are accessible through the Census Research Data Center (RDC). The public-use SIPP files only include state and metropolitan area identifiers (for pre-2004 panels) but lack county-level information.

conditions Lutz and Sand (2019). This county-level heterogeneity enables the construction of instrumental variables that capture more accurate exogenous variation, facilitating the identification of the causal effect of housing wealth.

In robustness checks, I employ alternative instrumental variables using different measures to approximate local housing supply elasticities at various geographic levels (MSA or state). These alternative specifications yield largely consistent results, though with somewhat lower statistical significance, reinforcing the value of the county-level approach in the primary analysis.

The underlying intuition of this instrument is that longer homeowner exposure to county-level housing market booms leads to greater housing wealth accumulation, where county-level housing market booms are caused by exogenous demand and supply market conditions—specifically, credit expansion in supply-constrained areas. As discussed in section 4, bank credit expansion increases mortgage credit availability, driving up housing demand. In areas where housing supply cannot easily expand due to geographic or regulatory constraints, this increased demand translates directly into higher housing prices rather than expanded housing stock. The relationship between demand and supply conditions and county housing market booms is consistent with the literature (Favara and Imbs, 2015; Yang, 2024b), which finds that bank deregulation raised housing prices significantly in counties with restricted housing supply.

At the household level, the timing of home purchase matters critically for housing wealth accumulation, even for households living in the same county. Earlier purchasers in deregulated, supply-constrained markets experience longer periods of price appreciation, accumulating substantially more housing wealth than later buyers in identical neighborhoods.

The strength of this instrument lies in its ability to capture both geographic and temporal variation, as demonstrated in the following scenarios:

- **Scenario 1: Geographic Variation**: Two households purchase homes in 1995—Household A in a deregulated county with inelastic housing supply, and Household B in a county with either no deregulation or elastic supply. Household A experiences greater housing wealth appreciation due to credit-driven price increases in supply-constrained markets.

- **Scenario 2: Temporal Variation**: Two households live in the same county where deregulation occurred in 1996, but Household A purchased in 1995 while Household B purchased in 1998. Despite identical locations, Household A accumulates more housing wealth through three additional years of credit expansion exposure.

The geographic variation captures differences in local housing markets across counties, identifying how similar households in different locations experience divergent housing wealth trajectories. The temporal variation captures differences in home purchase timing, distinguishing between otherwise similar households that bought homes at different points in the housing cycle. This household-level temporal variation provides key identification advantages. First, it addresses endogenous residential

sorting concerns by comparing households within the same geographic area who differ only in purchase timing, where the timing is largely determined by life-cycle factors rather than strategic decisions to maximize housing wealth appreciation.[18] This approach controls for time-invariant local characteristics while creating quasi-random variation in exposure duration to housing booms, where two equally capable households in the same county experience different housing wealth trajectories simply based on whether one purchased in different years. Second, by capturing variation at the individual household level, it enables more precise identification of causal effects and analysis of heterogeneous responses across different demographic characteristics.

This dual-source variation exploits differences in housing wealth accumulation that are plausibly exogenous to individual household characteristics or contemporaneous local economic conditions, allowing us to compare labor market decisions between workers with identical observable characteristics but different housing wealth due solely to location and purchase timing—factors unlikely to directly affect current labor market decisions.

The rich SIPP household-level data allows me to test whether the instrument influences non-housing wealth or borrowing. Later results show these effects are statistically insignificant, confirming that housing wealth, rather than general wealth effects or credit access, is the primary channel affecting labor market outcomes.

The analysis employs a two-stage least squares (2SLS) instrumental variables framework. The main specification estimates Equation (1) using the instrument Z_{ict} constructed in Equation (2). The first-stage regression for housing wealth takes the following form:

$$LogHW_{it} = \beta_1 Z_{ict} + \beta_2 X_{it} + \mu_c + \gamma_t + e_{ijt} \tag{3}$$

I cluster the standard errors at the level of the household to allow for arbitrary correlation within households and across time.

7 Main Results

7.1 First-Stage Results

Table 2 presents the first-stage regression results estimating the impact of an instrumental variable (IV) on housing equity across three model specifications. The IV exhibits a positive and statistically significant effect on housing equity in all specifications, with coefficients of 0.043, 0.038, and 0.039 (each significant at the 1% level). All models include state and year fixed effects, while the second and third specifications additionally incorporate individual and household controls. The third model

[18] A robustness check using expected home purchase timing instead of actual timing also helps alleviate concerns about the endogeneity of home purchase timing

further includes local controls and state-specific trends. The estimated effects are highly significant, with F-statistics ranging from 20 to 22, indicating that the instrument provides a strong source of variation in household housing wealth (Stock and Yogo, 2002). The analysis also explores gender heterogeneity (Columns 4–5), revealing statistically significant effects for both females (0.037) and males (0.041), with nearly identical magnitudes.

7.2 Second-Stage Results

Table 3 presents results analyzing the relationship between housing equity and labor force non-participation, comparing OLS and 2SLS estimation methods across the total sample and by gender. The key independent variable is log housing wealth. The OLS estimates show that a 1% increase in housing equity is associated with a 0.007 percentage point increase in the probability of not participating in the labor force for the total sample. This effect varies by gender: stronger for females (0.013 percentage points) and slightly negative for males (-0.003 percentage points). The 2SLS estimates, which address potential endogeneity, reveal larger effects: a 1% increase in housing equity corresponds to a 0.068 percentage point increase in labor force non-participation for the total sample. Again, gender differences emerge with a stronger effect for females (0.098 percentage points) and a negative but statistically insignificant effect for males (-0.040 percentage points). To contextualize these findings: given average household housing wealth of approximately $118,200, a $100,000 increase in housing equity is associated with a 5.78 percentage point increase in the probability of labor force non-participation for females. All analyses include comprehensive controls for state and year fixed effects, individual and household characteristics, local economic conditions, and state-specific trends. The efficient F-statistics for the 2SLS models confirm strong instrument relevance. Overall, the results suggest that increases in housing wealth reduce the labor force participation rate, particularly among female workers. These results align with prior literature demonstrating that labor supply decisions are responsive to changes in wealth, such as lottery winnings or unearned income (Imbens et al., 2001; Cesarini et al., 2017; Golosov et al., 2024) and housing wealth (Henley, 2004; Disney and Gathergood, 2018; Li et al., 2020). Particilalry, Golosov et al. (2024) find A lottery win of $100,000 causes a 6.4 percentage point increase in the likelihood of an older person exiting wage labor, and a 4.6 percentage point increase in the likelihood of a younger person exiting wage labor (where exit is defined as five years with zero wage labor income).which is inline with our results.

The literature on housing wealth's causal effects on labor supply is sparse, posing challenges for comparison but revealing demographic and contextual variations. In the UK, Disney and Gathergood (2018) find that a 10% house price increase reduces labor force participation by 1.3 percentage points and annual hours worked by 27 for married/cohabiting women under 40, with no significant effects for single or older women. Similarly, Henley (2004) reports that a 10% house price decline increases UK men's weekly work by 18 minutes (against a 39.7-hour mean). In China, Li et al. (2020) estimates that a 10,000-unit increase in housing wealth reduces annual earnings by 203.06 (5.97% of the control

mean).

8 Robustness Checks

8.1 Tests on Exclusion Restriction

Table 4 reports exclusion restriction tests, a critical requirement for the IV approach's validity. This condition requires that the instrument affects the outcome only through housing wealth, not through alternative channels. We evaluate this by regressing the IV on three potential confounders: (1) housing debt, (2) other debt, (3) non-housing wealth, and (4) household income while controlling for fixed effects and covariates.

The results strongly support the exclusion restriction. For housing debt, the IV coefficient is -0.008 (SE=0.008), showing an insignificant negative relationship. Other debt displays an even smaller, statistically insignificant coefficient (0.001, SE=0.006). Non-housing wealth shows a marginally positive but insignificant association (0.008, SE=0.008). Household income is positive and also insignificant (0.002, SE=0.004). The consistently null results across all three tests (housing debt, other debt, and non-housing wealth) support the instrument's validity. The absence of statistically significant relationships suggests the IV operates primarily through housing wealth rather than alternative channels, thereby satisfying the exclusion restriction. This evidence strengthens the causal interpretation of the main results regarding housing equity's effect on labor force participation.

Additionally, in Appendix Table A1, I examine how deregulation affects housing and labor market outcomes at the county level, as in Yang (2024b). Column (1) of Table A1 reports the impact on county-level FHFA house price index, showing that deregulation leads to significantly higher house prices, with the coefficient significant at the 1 percent level.[19] Columns (2) through (5) examine the impact of deregulation on local labor market outcomes, using county-level data from the QCEW to measure growth in employment and wages. Columns (2) and (3) consider all industries, while columns (4) and (5) focus on female-dominated industries such as education and health services, leisure and hospitality, and financial industries. Notably, the results show no significant relationship between deregulation and labor market outcomes, regardless of industry.[20]

The results in this section provide strong evidence supporting the instrumental variable's validity.

[19]This effect is more pronounced and significant in areas with less land availability (0.042 and 0.05). These findings are in line with those of Favara and Imbs (2015). Following Favara and Imbs (2015), I also use the Home Mortgage Disclosure Act (HMDA) database to test the impact of the banking deregulation on the number of mortgage loans originating at the county level and confirm the positive relationship between deregulation and loan volume. These results are available upon request.

[20]These findings align with Célerier and Matray (2019), who also noted that bank deregulation in the 1990s had a relatively limited impact on income and economic growth. While earlier studies documented the effects of bank deregulation on income and economic growth in the 1970s and 1980s, this paper and Célerier and Matray (2019) focus on 1990s deregulation when the financial market was more integrated, potentially explaining the differing results.

By showing no significant correlation between the IV and alternative channels, these findings strengthen the causal interpretation of the housing equity effects on labor force participation. This validation analysis is critical for establishing the robustness of the study's main findings.

8.2 Tests on Different Specifications

Table 5 presents robustness checks examining the relationship between housing equity and labor force participation across alternative specifications, with results reported for the full sample and by gender.

Alternative IV Specifications: The first set of tests addresses potential concerns about instrument construction. Alternative IV 1 tightens the land supply constraint by using a 60% (rather than the baseline 70%) threshold for developable land, yielding stable results with significant effects for the full sample (0.051) and females (0.081) but not males. Alternative IV 2 employs the continuous proportion of developable land instead of a binary cutoff, producing smaller coefficients with only marginal significance for females (0.043). Alternative IV 3 replaces household-level purchase years with state-cohort averages to mitigate endogeneity concerns; the estimates (0.033 for the full sample, 0.049 for females) align with the baseline, suggesting limited bias from endogenous timing of home purchases.[21]

Alternative Model Specifications: The second set of tests evaluates the sensitivity of results to model specifications. Adding household controls (e.g., housing debt, non-housing wealth) increases the coefficients (0.073 for the full sample, 0.098 for females), indicating that omitted household variables do not drive the baseline results. Incorporating MSA fixed effects produces the largest estimates (0.098 for the full sample, 0.148 for females), though with a modest sample reduction, implying that local labor market heterogeneity strengthens the housing-wealth relationship. Using housing equity in levels (rather than logs) preserves significance (0.051 for the full sample, 0.072 for females), supporting the baseline functional form. All models include state and year fixed effects, individual and household controls, local economic controls, and state-specific trends. The stability of results across specifications underscores the robustness of the main findings.

8.3 Other Labor Market Outcomes

Table A2 presents the results of an analysis examining the impact of housing wealth on various labor market outcomes using SIPP. The outcomes examined include Hours Worked, Part-Time employment, Hourly Wage, Unemployment, and Household Income. The results suggest that an increase in housing

[21] I also examine other instrumental variable (IV) constructions. These include: (1) using state-level banking deregulation exposure rather than household-level measures, (2) measuring land availability with state-level rather than county-level data, and (3) employing MSA-level housing supply elasticity measures (the Wharton Residential Land Use Regulatory Index (WRLURI) from Gyourko et al. (2008) and the proportion of undevelopable land from Saiz (2010).). The results become less statistically significant with these alternative IV specifications, underscoring the importance of using household-level IV construction and county-level housing supply measures for capturing the relevant wealth effects.

equity is associated with a decrease in hours worked (-3.972), though this effect is not statistically significant at conventional levels. The impact on part-time employment is negligible (0.000). There appears to be a positive effect on hourly wages (2.562), but again, this is not statistically significant. Interestingly, there is a small positive association with unemployment (0.015), and a larger positive effect on household income (5.544), though neither of these effects reaches statistical significance. The analysis employs a robust set of controls, including state and year fixed effects, individual and household controls, local economic controls, and state-specific trends. The efficient F-statistics, ranging from 20.75 to 21.13, indicate strong instrument relevance in the instrumental variables approach. Overall, while the directions of the effects are noteworthy, the lack of statistical significance suggests that the relationship between housing equity and these broader labor market outcomes may be more nuanced or weaker than its relationship with labor force participation.

9 Discussion of Mechanisms

In this section, I design and conduct empirical tests to examine the housing wealth effects on labor supply reductions, with a particular focus on gender differences. We first provide two pieces of direct evidence that these effects are largely driven by childcare responsibilities. First, the results are statistically significant only among females with young children, consistent with the greater childcare demands in early childhood. Second, exploiting a unique survey question on reasons for not working (asked among those out of the labor force), I find that higher housing wealth significantly increases the probability of non-participation specifically for child-related reasons, but shows no significant association with non-child-related reasons. Together, these findings highlight the central role of childcare responsibilities in mediating the impact of housing wealth on labor supply.

Building on evidence that childcare responsibilities are the primary driver of the housing wealth effect on labor supply, we conduct further heterogeneity analyses across several key dimensions: the woman's own education level, her spouse's education level, marital status, and household financial conditions (e.g., non-housing/liquid wealth). These tests probe additional mechanisms through which housing wealth influences labor force non-participation for child-related reasons, including commitment or bargaining effects, precautionary motives, liquidity constraints, and preferences for childcare quality.

9.1 Empirical Tests of the Mechanisms of Childcare Responsibility

Table 6 presents heterogeneous effects of log housing equity on labor force participation, stratified by the presence and age of the youngest child in the household. For households without children, the estimates are small and statistically insignificant across all groups (0.040 for the full sample, 0.070 for females, and 0.011 for males). In contrast, households with young children (youngest child $\leq$ 6 years

old) exhibit strong and statistically significant positive effects for the full sample (0.089, $p < 0.01$) and especially for females (0.136, $p < 0.01$), while the effect for males remains small and insignificant (0.034). For households with older children (youngest child > 6 years old), the point estimates are close to zero or slightly negative and statistically insignificant (-0.009 for the full sample, -0.033 for females, and 0.044 for males).

These patterns indicate that the positive housing wealth effects on labor force participation, interpreted here as reduced labor supply among women, are highly concentrated among mothers with very young children, consistent with a childcare-responsibility or commitment-device mechanism. First-stage F-statistics are strong across all specifications (ranging from 15.57 to 22.97), supporting instrument validity. Additional unreported results show that the effect is particularly pronounced among college-educated mothers with young children, while it remains modest (and often insignificant) among non-college-educated mothers. The magnitude is also larger for mothers whose youngest child is under 3 years old compared to those with children aged 3–6.[22]

Table 7 presents evidence on the mechanisms underlying the housing wealth effect on labor force non-participation, distinguishing between child-related and non-child-related reasons for being out of the labor force.[23] The results show that higher log housing equity significantly increases the probability of non-participation, specifically due to child-related reasons, with coefficients of 0.042 ($p < 0.01$) for the full sample and 0.067 ($p < 0.01$) for females, while the estimate for males is small and insignificant (0.016). In contrast, the effect on non-child-related reasons for non-participation is smaller and statistically insignificant across all groups (0.031 for the total sample, 0.035 for females, and 0.027 for males). First-stage F-statistics are strong in all specifications (ranging from 21.24 to 24.72), confirming instrument relevance. These patterns provide direct support for the hypothesis that housing wealth gains primarily reduce women's labor supply through increased childcare responsibilities rather than through general income effects, precautionary motives, or other non-child-related channels.

9.2 Discussion of Other Mechanisms

Results regarding the discussion of other mechanisms are presented in Table 8. The effect is substantially stronger among college-educated mothers (0.116 for females, p<0.01p) than among non-college-educated mothers (0.043, insignificant). Since college-educated women typically exhibit lower labor supply income elasticities—meaning their labor decisions are less sensitive to income changes—this

[22]Supplementary analyses find no statistically significant effects of housing wealth on demographic outcomes (e.g., marriage rates, fertility) or other financial outcomes (e.g., loan amounts, non-housing assets). These null results help rule out alternative mechanisms. Full tables are available upon request.

[23]Among respondents who exit the labor force during the sample period, 56% cite childcare responsibilities as their primary reason for not working. This aggregate figure masks substantial gender differences: 66% of non-working women report childcare as the main reason, while 12% cite health-related reasons and the remainder cite retirement, schooling, market conditions, or leisure. In contrast, only 8% of non-working men cite childcare, while 49% report health-related reasons. These stark gender differences in exit motivations reinforce that housing wealth shocks affect female labor supply primarily through the childcare channel rather than through general income effects, early retirement, or increased leisure consumption that would affect both genders more symmetrically.

pattern suggests the response is unlikely to reflect a standard income effect. Panel B further shows that the effect is more pronounced in households where both parents hold college degrees (0.116 for females, p<0.01) compared to households where only one or neither parent has a college degree (small and insignificant estimates).[24]

The patterns in Panels A and B suggest that highly educated mothers with highly educated spouses invest more in children's early education and respond more to housing wealth changes. This differential also argues against precautionary saving as the main channel, as dual-college-educated households generally have higher and more stable incomes, reducing the need for precautionary buffers. Instead, these findings align with a mechanism in which dual-college education signals stronger valuation of high-quality childcare and child-rearing time, leading women in such households to reduce labor supply more readily when housing wealth increases, thereby reallocating time toward childcare.

Panel C shows that the effects are substantially stronger among married respondents (0.088 for females, $p < 0.01$) than among never-married ones (noisy and mostly insignificant estimates). The weaker results for never-married individuals likely reflect both smaller sample sizes and low first-stage F-statistics, which limit statistical power. However, the marriage differential is consistent with commitment effects or intra-household bargaining mechanisms that facilitate labor supply adjustments. Married women may have greater flexibility to reduce work hours due to shared financial security with a spouse, or because marriage enables household specialization in response to wealth shocks. These interpretations should be treated cautiously, as marriage is not randomly assigned and may correlate with unobserved preferences for traditional gender roles or household arrangements that independently affect labor supply responses.[25]

Finally, Panel D reveals that the effect is larger in low non-housing wealth households (0.110 for females, p<0.01) than in high non-housing wealth households (0.068 for females, insignificant), supporting a role for liquidity constraints in amplifying the response. Families with fewer financial buffers beyond housing wealth may be more responsive to housing wealth appreciation, as it represents a larger share of their total assets and potentially relaxes binding constraints on consumption smoothing or childcare investments. Across all subsamples in Panels C and D, male estimates remain small and insignificant, reinforcing the highly gendered nature of the childcare channel. Importantly, further breakdowns within these groups show that the presence of young children remains critical for observing negative female labor supply effects. This suggests that neither marriage status nor wealth constraints alone can fully explain the housing wealth effect on female labor supply—rather, these factors interact

[24]Further breakdowns by parental education combinations (college mother/non-college father, non-college mother/college father, or both non-college) show no significant effects. Results by spouse wage (above vs. below median) show stronger effects for higher-wage spouses. Across all subgroups, the presence of young children remains important for negative labor supply effects. These results are available upon request.

[25]An alternative explanation is that married women face stronger social or household norms regarding childcare responsibilities, which could amplify their labor supply response to housing wealth changes independent of bargaining or commitment mechanisms.

with childcare demands to shape women's labor market responses.[26]

10 Conclusion

This study examines the causal relationship between housing wealth and labor supply using confidential Survey of Income and Program Participation (SIPP) data with restricted geographic information. The research offers two key contributions to economic literature. First, it introduces a methodological innovation through the development of a novel household-level instrument that isolates housing wealth effects. This instrument captures variation from geographic differences in housing price growth driven by credit expansion, local housing supply constraints, and individual households' home purchase timing decisions. Second, the research identifies specific mechanisms through which housing wealth affects labor decisions.

The findings establish that housing wealth exhibits a significant negative correlation with labor supply, and this relationship is strongly gender-differentiated. For women, a 1% increase in home equity raises the probability of labor force exit by 0.098 percentage points, while for men, no statistically significant effect was detected. College-educated mothers with young children demonstrate the strongest response to changes in housing wealth. This pattern suggests housing wealth functions as a "commitment device" - an understudied mechanism whereby increased home equity provides financial flexibility that enables certain household members, particularly mothers, to reduce labor market participation to address family responsibilities.

In the broader economic context, this research enhances our understanding of how housing market volatility, especially during the significant boom-bust cycle of recent decades, transmits to labor markets. It complements existing research on the large literature on the persistent gender gaps (Goldin, 2014; Blau and Kahn, 2017; Kleven et al., 2019) by identifying housing wealth fluctuations as a previously unexamined factor contributing to gendered labor market participation patterns. These insights have important implications for policymakers who navigate the dynamics of the housing market and their downstream effects on labor markets and gender inequality.

[26]Table A3 reports further subsample heterogeneity in the effect of log housing equity on the probability of being out of the labor force, focusing on race/ethnicity (white vs. non-white) and geographic location (metropolitan vs. non-metropolitan areas). The estimated effects are consistently larger in magnitude and more statistically significant among white individuals compared to non-white individuals. The metropolitan/non-metropolitan split shows significant positive effects only in metropolitan areas, with especially strong results for females. Additionally, the relationship appears more pronounced among mothers of middle age and in the post-2000 subsample. Those results are available upon request.

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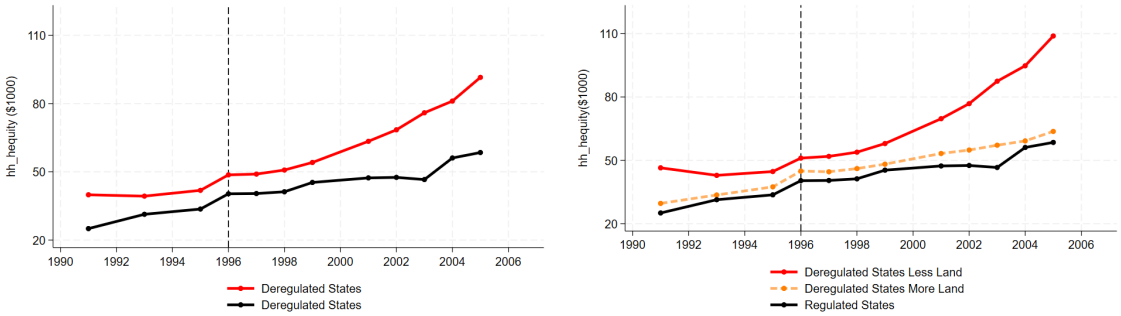
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Figures and Tables

Figure 1: Household Housing Wealth in Deregulated vs. Regulated States in SIPP



(a) HH Housing Wealth (b) HH Housing Wealth

Notes: The sample is constructed from the public-use versions of the 1991–2004 Survey of Income and Program Participation (SIPP) panels, limited to individuals aged 15–55 who are homeowners, with observations covering the 1991–2005 period. Less land measures whether the state-level developable land is less than 70 percent of the total area based on satellite data collected by Lutz and Sand (2019).

Table 1: Summary Statistics of the SIPP (1991-2005)

	(1) Total	(2) Female	(3) Male	(4) Diff.
Individual Variables				
Age	40.98	40.65	41.34	0.63***
Female	0.52	1.00	0.00	-1.00
Married	0.84	0.83	0.84	0.03***
Num of Children	1.20	1.20	1.20	0.01
White	0.88	0.87	0.89	0.02***
Black	0.07	0.08	0.07	-0.01***
Other Race	0.05	0.05	0.05	-0.00***
Hispanic	0.06	0.06	0.06	-0.00*
Less than HS	0.06	0.06	0.06	0.01***
High School	0.25	0.25	0.25	0.00
College	0.57	0.58	0.55	-0.03***
Graduate	0.12	0.12	0.13	0.02***
Not in LF	0.18	0.27	0.08	-0.19***
Hours Worked (Week)	40.64	37.11	44.07	6.98***
Hours Worked (Month)	176.4	161.1	191.3	30.20***
Hourly Wage	24.95	21.31	28.48	7.27***
Unemployed	0.08	0.08	0.08	-0.00***
Household Variables				
House Purchase Year	1996	1996	1996	0.17***
House Tenure	6.38	6.45	6.30	-0.17***
HH Income ($1000)	92.54	91.34	93.86	3.06***
Housing Equity ($1000)	118.2	118.8	117.5	-0.80
Housing Debt ($1000)	112.3	110.7	114	3.64***
Non-Housing Equity ($1000)	93.37	93.00	93.77	1.63**
Non-Housing Debt ($1000)	31.41	30.83	32.05	1.38***
Metro Area	0.83	0.83	0.83	-0.00
Local Labor Market Variables				
QCEW Employment	7594	7609	7577	-63.38
QCEW Wage	46350	46380	46310	-76.50
Observations	146000	77000	69000	

Notes: The sample is constructed from the restricted-use versions of the 1991–2004 Survey of Income and Program Participation (SIPP) panels, limited to individuals aged 15–55 who are homeowners, with observations covering the 1991–2005 period. All monetary variables are inflation-adjusted to 2010 dollars.

Table 2: Housing Equity and Not in Labor Force: First Stage

	Log Housing Equity				
		Total		Female	Male
	(1)	(2)	(3)	(4)	(5)
IV	0.043***	0.038***	0.039***	0.037***	0.041***
	(0.010)	(0.009)	(0.008)	(0.008)	(0.008)
N	146000	146000	146000	77000	69000
R^2	0.102	0.193	0.199	0.201	0.199
State FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Ind and HH Controls	No	Yes	Yes	Yes	Yes
Local Controls	No	No	Yes	Yes	Yes
State Trends	No	No	Yes	Yes	Yes

Notes: The sample is constructed from the restricted-use versions of the 1991–2004 SIPP panels, limited to individuals aged 15–55 who are homeowners, with observations covering the 1991–2005 period. The IV is constructed as the measure of households' duration of exposure to housing market booms driven by credit expansion in housing supply-constrained areas. Standard errors are presented in parentheses and are clustered at the county level. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table 3: Housing Equity and Not in Labor Force: OLS and 2SLS

	OLS			2SLS		
	Total	Female	Male	Total	Female	Male
	(1)	(2)	(3)	(4)	(5)	(6)
Log Housing Equity	0.007***	0.013***	-0.003***	0.069**	0.098**	0.040
	(0.001)	(0.002)	(0.001)	(0.028)	(0.039)	(0.029)
N	146000	77000	69000	146000	77000	69000
R^2	0.092	0.068	0.025	0.058	0.022	0.022
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Ind and HH Controls	Yes	Yes	Yes	Yes	Yes	Yes
Local Controls	Yes	Yes	Yes	Yes	Yes	Yes
State Trends	Yes	Yes	Yes	Yes	Yes	Yes
Efficient F				21.13	19.72	22.29

Notes: The sample is constructed from the restricted-use versions of the 1991–2004 SIPP panels, limited to individuals aged 15–55 who are homeowners, with observations covering the 1991–2005 period. The IV is constructed as the measure of households' duration of exposure to housing market booms driven by credit expansion in housing supply-constrained areas. Standard errors are presented in parentheses and are clustered at the county level. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table 4: Test of Exclusion Restriction: Exclude Other Channels

	Log Housing Debt	Log Other Debt	Log Non-Housing Equity	Log Income
	Total	Total	Total	Total
	(1)	(2)	(3)	(4)
IV	-0.008	0.001	0.008	0.002
	(0.008)	(0.006)	(0.008)	(0.004)
N	146000	146000	146000	146000
R^2	0.143	0.046	0.198	0.044
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Ind and HH Controls	Yes	Yes	Yes	Yes
Local Controls	Yes	Yes	Yes	Yes
State Trends	Yes	Yes	Yes	Yes

Notes: The sample is constructed from the restricted-use versions of the 1991–2004 SIPP panels, limited to individuals aged 15–55 who are homeowners, with observations covering the 1991–2005 period. The IV is constructed as the measure of households' duration of exposure to housing market booms driven by credit expansion in housing supply-constrained areas. Outcome variables are in $1,000 and natural log. Standard errors are presented in parentheses and are clustered at the county level. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table 5: Robustness Checks: Different Specifications

	Alternative IV 1			Alternative IV 2			Alternative IV 3		
	Total	Female	Male	Total	Female	Male	Total	Female	Male
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Log Housing Equity	0.051**	0.081**	0.019	0.021	0.043*	-0.000	0.033**	0.049**	0.015
	(0.022)	(0.033)	(0.022)	(0.018)	(0.026)	(0.015)	(0.016)	(0.024)	(0.016)
N	146000	77000	69000	146000	77000	69000	146000	77000	69000
Efficient F	15.75	15.55	16.75	11.13	11.22	10.12	19.78	18.35	18.75
	More Controls			Add MSA dummies			Level of Housing Equity		
	Total	Female	Male	Total	Female	Male	Total	Female	Male
	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Log Housing Equity	0.073**	0.098**	0.047	0.098***	0.148***	0.045	0.051**	0.072**	0.029
	(0.028)	(0.039)	(0.032)	(0.036)	(0.049)	(0.032)	(0.021)	(0.030)	(0.021)
N	146000	77000	69000	116000	61500	55000	146000	77000	69000
Efficient F	14.13	16.22	15.65	14.32	14.21	13.65	17.55	16.21	15.77
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind and HH Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Local Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State Trends	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The sample is constructed from the restricted-use versions of the 1991–2004 SIPP panels, limited to individuals aged 15–55 who are homeowners, with observations covering the 1991–2005 period. The IV is constructed as the measure of households' duration of exposure to housing market booms driven by credit expansion in housing supply-constrained areas. Standard errors are presented in parentheses and are clustered at the county level. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table 6: Housing Equity and Not in Labor Force: By Age of Children
Mechanisms Test I

	No children			Youngest Children <=6			Youngest children > 6		
	Total	Female	Male	Total	Female	Male	Total	Female	Male
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Log Housing Equity	0.040	0.070	0.011	0.089**	0.136**	0.034	-0.009	-0.033	0.044
	(0.046)	(0.074)	(0.050)	(0.038)	(0.062)	(0.024)	(0.050)	(0.084)	(0.036)
N	55550	29000	26500	41500	21500	20000	49000	26500	22500
Efficient F	19.27	21.25	15.57	20.22	19.28	20.62	21.93	20.33	22.97
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind and HH Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Local Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State Trends	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The sample is constructed from the restricted-use versions of the 1991–2004 SIPP panels, limited to individuals aged 15–55 who are homeowners, with observations covering the 1991–2005 period. The IV is constructed as the measure of households' duration of exposure to housing market booms driven by credit expansion in housing supply-constrained areas. Standard errors are presented in parentheses and are clustered at the county level. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table 7: Housing Equity and Why not Working:
Mechanisms Test II

	Child-related Reason			Non-Child Related Reason		
	Total	Female	Male	Total	Female	Male
	(1)	(2)	(3)	(4)	(5)	(6)
Log Housing Equity	0.042**	0.067**	0.016	0.031	0.035	0.027
	(0.018)	(0.033)	(0.013)	(0.020)	(0.023)	(0.022)
N	140,000	73,500	66,000	140,000	73,500	66,000
Efficient F	23.16	21.24	24.72	23.16	21.24	24.72
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Ind + HH FE	Yes	Yes	Yes	Yes	Yes	Yes
Local controls	Yes	Yes	Yes	Yes	Yes	Yes
State trends	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The sample is constructed from the restricted-use versions of the 1991–2004 SIPP panels, limited to individuals aged 15–55 who are homeowners, with observations covering the 1991–2005 period. The IV is constructed as the measure of households' duration of exposure to housing market booms driven by credit expansion in housing supply-constrained areas. Columns (1)–(3) use a binary dependent variable equal to 1 if the individual is out of the labor force due to child-related reasons, and 0 otherwise (including in the labor force). Columns (4)–(6) use an analogous binary indicator equal to 1 if the individual is out of the labor force due to non-child-related reasons, and 0 otherwise. Non-child-related reasons include health issues, education, caregiving for elderly or disabled relatives, retirement, and other non-child-related explanations. Standard errors are presented in parentheses and are clustered at the county level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 8: Housing Equity and Not in Labor Force:
Mechanisms Test III

Panel A	Mother College			Mother Non-college		
	Total	Female	Male	Total	Female	Male
	(1)	(2)	(3)	(4)	(5)	(6)
Log Housing Equity	0.075***	0.116***	0.029	0.052	0.043	0.062
	(0.028)	(0.041)	(0.026)	(0.051)	(0.073)	(0.060)
N	99500	52500	47000	46500	24000	22000
Efficient F	19.16	18.18	18.8	17.22	13.82	18.04
Panel B	Both Parents College			Other		
	Total	Female	Male	Total	Female	Male
Log Housing Equity	0.080**	0.106**	0.049	0.011	-0.023	0.057
	(0.033)	(0.052)	(0.031)	(0.049)	(0.076)	(0.058)
N	63,500	32,000	32,000	49,000	24,500	24,500
Efficient F	19.36	19.73	18.81	15.96	14.78	17.37
Panel C	Married			Never-married		
	Total	Female	Male	Total	Female	Male
Log Housing Equity	0.070**	0.088**	0.051	0.043	0.115	0.000
	(0.031)	(0.044)	(0.032)	(0.067)	(0.133)	(0.000)
N	122000	63000	59000	9600	4600	5000
Efficient F	22.43	20.99	23.78	14.68	4.719	19.61
Panel D	High Non-Housing Wealth			Low Non-Housing Wealth		
	Total	Female	Male	Total	Female	Male
Log Housing Equity	0.074*	0.068	0.075*	0.069*	0.110**	0.033
	(0.040)	(0.062)	(0.041)	(0.036)	(0.053)	(0.038)
N	64,500	33,500	31,000	57,500	29,500	28,000
Efficient F	17.99	18.15	17.52	27.71	25.24	29.79
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Ind and HH Controls	Yes	Yes	Yes	Yes	Yes	Yes
Local Controls	Yes	Yes	Yes	Yes	Yes	Yes
State Trends	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The sample is constructed from the restricted-use versions of the 1991–2004 SIPP panels, limited to individuals aged 15–55 who are homeowners, with observations covering the 1991–2005 period. The IV is constructed as the measure of households' duration of exposure to housing market booms driven by credit expansion in housing supply-constrained areas. Non-housing wealth is classified as high or low based on whether it exceeds or falls below the sample median. Standard errors are presented in parentheses and are clustered at the county level. *** p < 0.01, ** p < 0.05, * p < 0.10.

Online Appendix to:
"Housing Wealth and Labor Supply: Evidence from Geographically-linked Microdata"

Xi Yang

January, 2026

Appendix: Interstate Bank Branching Deregulation in the U.S.

Most U.S. states historically restricted interstate banking and branching, dating back to colonial times.[27] In the 1970s and 1980s, the banking sector went through decades of deregulatory changes regarding banks' geographic expansion. However, interstate bank branching was still not allowed until 1994 (Calomiris, 2000; Kroszner and Strahan, 1999, 2014). In that year, the Interstate Banking and Branching Efficiency Act (IBBEA) was adopted, permitting bank holding companies to enter other states and operate branches. The IBBEA also granted individual states some latitude in deciding the timing of deregulation independently (Rice and Strahan, 2010; Favara and Imbs, 2015).

Figure A1 maps the timing of deregulation across states, with darker shades representing earlier deregulation. Most of the policy changes took place between 1996 and 2002. By 2005, eight mid-western states still were not deregulated, and we did not observe additional deregulation afterward. Deregulation is more common and happened earlier in eastern and western states and relatively uncommon or happened later among states in the middle. The cross-state and -year policy variation enables a staggered DID design to identify the causal effect of credit expansion. To capture this policy variation, a deregulation indicator is created to indicate whether the deregulation has taken place in a state. As in Rice and Strahan (2010), every state is assumed to be fully restricted in 1994, and deregulation is defined if the state has relaxed at least one restriction on interstate branching.[28]

Bank deregulation is a positive shock to local credit expansion, thus, an important driver of housing wealth among homeowners, suggesting that these instruments are relevant. However, the validity of those instruments also requires deregulations being an exogenous trigger of house prices with no direct effect on other factors such as local labor market conditions (the exclusion restriction

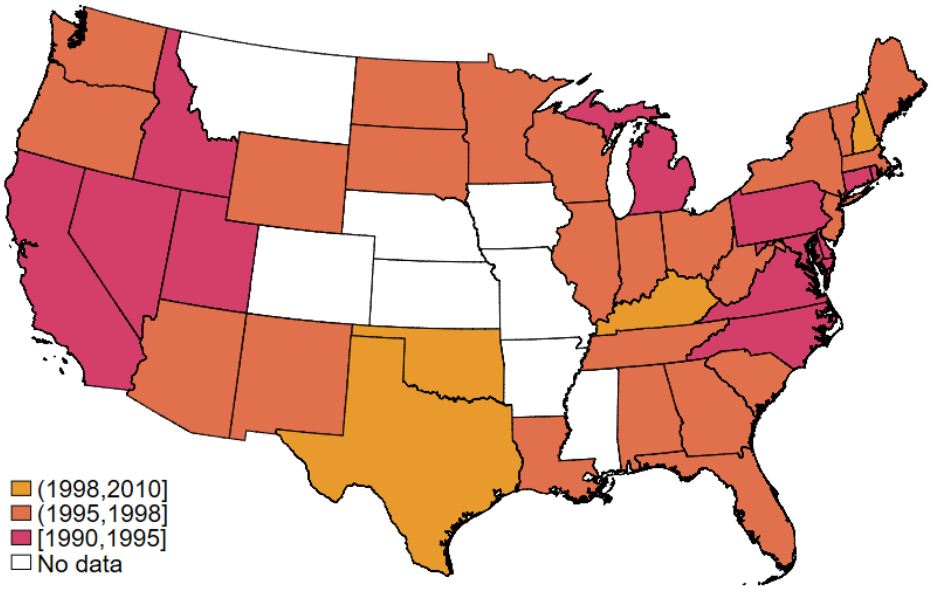
[27] Interstate banking is when bank-holding companies own banks in different states, while interstate branching is when a single bank has branches in multiple states without separate structures.

[28] The IBBEA permitted states to establish restrictions on out-of-state bank entry in four key areas: (i) setting a minimum age requirement for the targeted bank, (ii) prohibiting de-novo branching without explicit approval from state authorities, (iii) allowing the acquisition of individual branches without the need to acquire the entire bank, and (iv) implementing a state-wide deposit cap. Rice and Strahan (2010) compute a time-varying regulation index that ranges from 0 to 4 to capture the state-level branching restrictions. Previous studies have adopted this index to evaluate the impact of banking deregulation on the price of housing Favara and Imbs (2015) and financial inclusion Célerier and Matray (2019).

condition). The banking literature on the political economy of banking deregulations supports this assumption, which finds that banking deregulations reflect large banks' strength and political clout relative to small banks and are exogenous to local labor market conditions (Kroszner and Strahan, 2014). Meanwhile, several recent empirical exercises have made use of banking deregulations as instruments in other contexts, adding to the validity of our identifying assumptions (Favara and Imbs, 2015; Célerier and Matray, 2019).

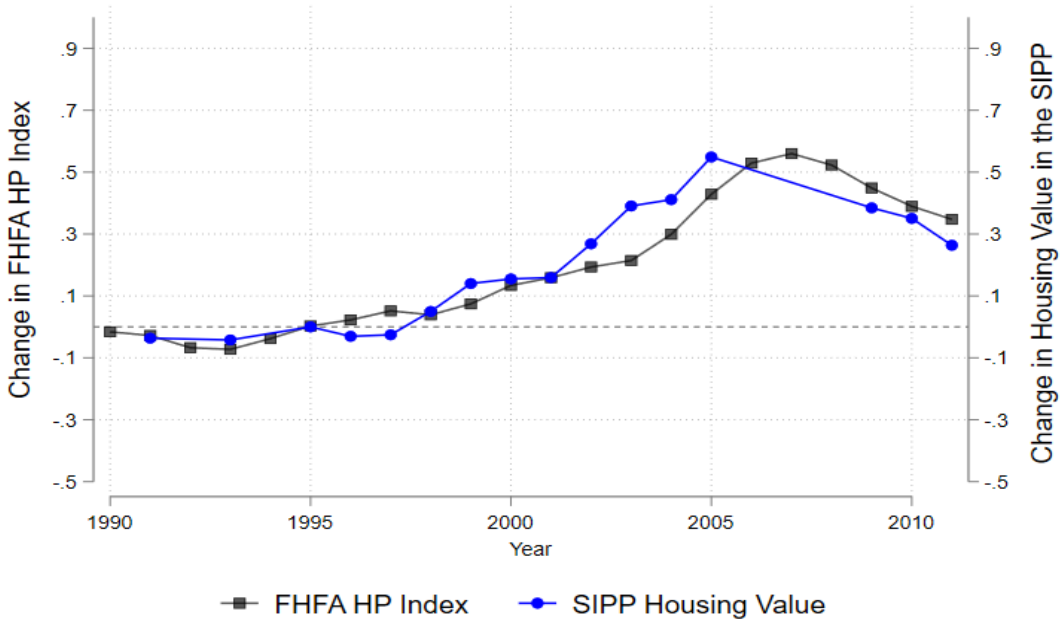
Appendix: More Figures and Tables

Figure A1: Timing of Interstate Bank Branching Deregulation



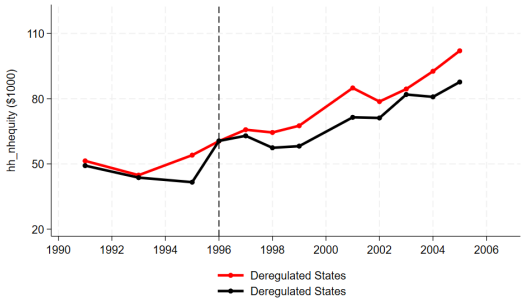
Notes: Data on interstate bank branching deregulation by state and by year come from Rice and Strahan (2010). Eight states never deregulated which include Arkansas, Colorado, Iowa, Kansas, Mississippi, Missouri, Montana, and Nebraska.

Figure A2: Housing Price Dynamics in FHFA and SIPP

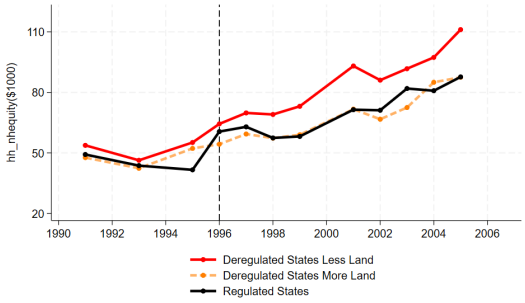


Notes: The housing price index comes from the Federal Housing Finance Agency (FHFA). In the SIPP, housing value is self-reported and collected in the Topical Module.

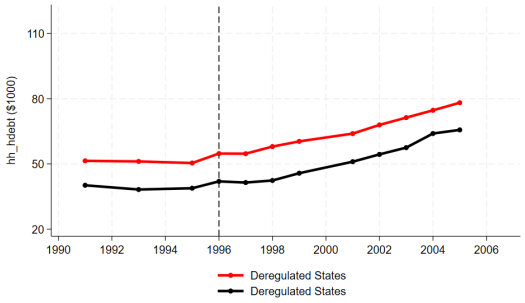
Figure A3: Other Household Outcomes in Deregulated vs. Regulated States



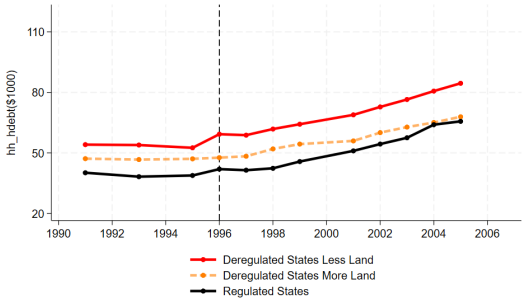
(a) Non-Housing Wealth



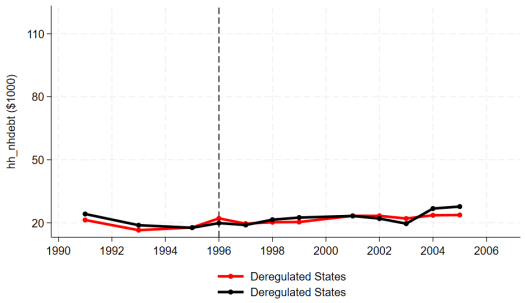
(b) Non-Housing Wealth



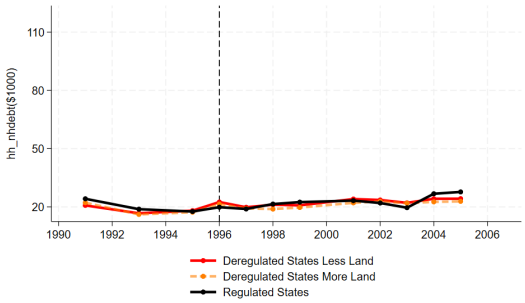
(c) Housing Debt



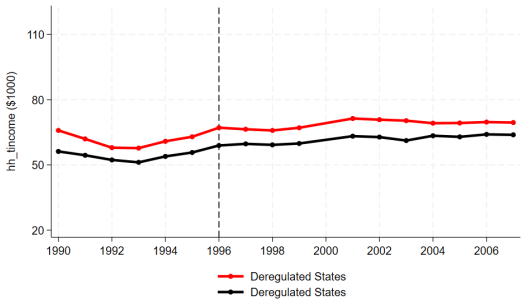
(d) Housing Debt



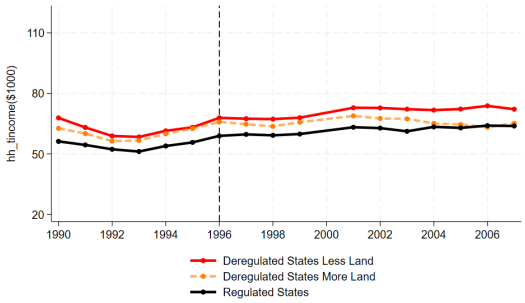
(e) Non-Housing Debt



(f) Non-Housing Debt



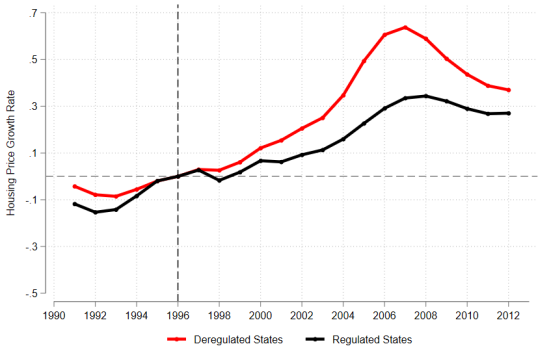
(g) HH Income



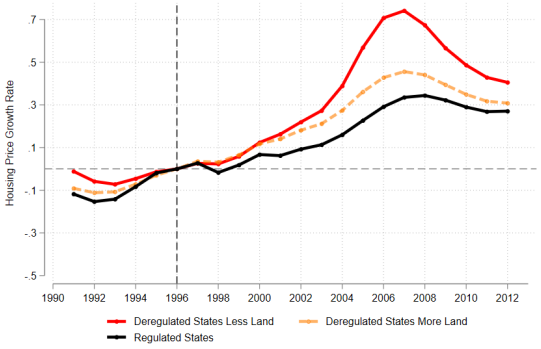
(h) HH Income

*Notes:*The sample is constructed from the public-use versions of the 1991–2004 Survey of Income and Program Participation (SIPP) panels, limited to individuals aged 15–55 who are homeowners, with observations covering the 1991–2005 period. Less land measures whether the state-level developable land is less than 70 percent of the total area based on satellite data collected by Lutz and Sand (2019).

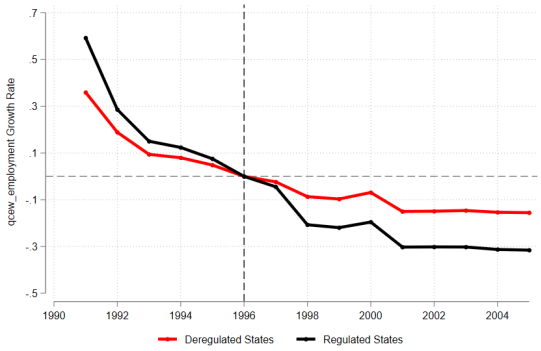
Figure A4: Housing Price, Employment, and Wage Trends in Deregulated vs. Regulated States



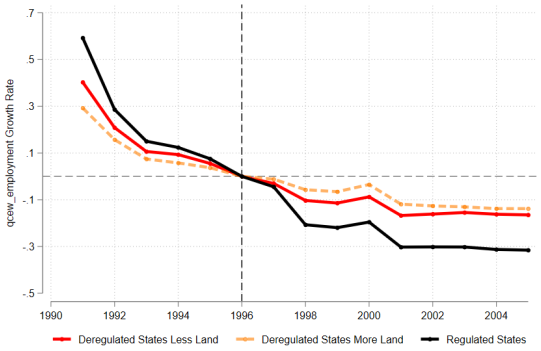
(a) Housing Price Growth Rate



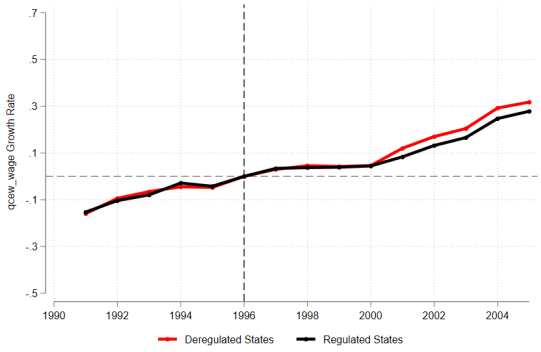
(b) Housing Price Growth Rate



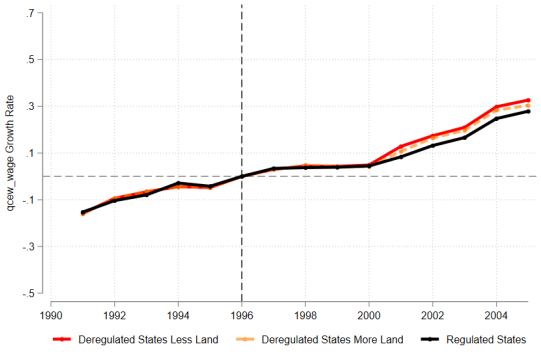
(c) QCEW Employment Growth Rate



(d) QCEW Employment Growth Rate



(e) QCEW Wage Growth Rate



(f) QCEW Wage Growth Rate

Notes: House price growth comes from the Federal Housing Finance Agency Housing Price Index (HPI). Employment and wage growth come from the Bureau of Labor Statistics Quarterly Census of Employment and Wages (QCEW) which provides information on employment and wages reported by employers covering more than 95 percent of U.S. jobs.

Table A1: Mechanism Test: Effects of Bank Branching Deregulation on County-Level Housing and Labor Market Outcomes

	FHFA	All Industries		Female-Dominated Industries	
	House Price	Employment	Wage	Employment	Wage
	(1)	(2)	(3)	(4)	(5)
Deregulation Dummy	0.041**	0.008	-0.004	0.012	-0.004
	(0.020)	(0.005)	(0.005)	(0.017)	(0.005)
Observations	5671	6556	6556	5552	5552
County FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Economic Controls	Yes	Yes	Yes	Yes	Yes
Demographic Controls	Yes	Yes	Yes	Yes	Yes

Notes: The outcome variable in column (1) is the log change in the FHFA house price index at the county level. The outcome in panels (2)-(5) are log changes of county-level employment and wage calculated based on the Quarterly Census of Employment and Wages (QCEW). Female-dominated industries include education and health services, leisure and hospitality, and financial industries according to the Bureau of Labor Statistics data in 2006. The deregulation dummy indicates whether the state has implemented the interstate bank branching deregulation. Each regression adopts the CSDID model. Standard errors are clustered at the state level. * p<0.10, ** p<0.05, *** p<0.01.

Table A2: Housing Equity and Other Labor Market Outcomes

	Hours Worked	Part Time	Hourly Wage	Unemployed	HH Income
	(1)	(2)	(3)	(4)	(5)
Log Housing Equity	-3.972	0.000	2.562	0.015	5.544
	(2.610)	(0.015)	(1.997)	(0.019)	(6.641)
N	114,000	114,000	114,000	146,000	146,000
Efficient F-stat	20.75	20.75	20.75	20.75	21.13
State FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Ind & HH Controls	Yes	Yes	Yes	Yes	Yes
Local Controls	Yes	Yes	Yes	Yes	Yes
State Trends	Yes	Yes	Yes	Yes	Yes

Notes: The sample is constructed from the restricted-use versions of the 1991–2004 SIPP panels, limited to individuals aged 15–55 who are homeowners, with observations covering the 1991–2005 period. The IV is constructed as the measure of households' duration of exposure to housing market booms driven by credit expansion in housing supply-constrained areas. Hours worked are measured per month. Standard errors are presented in parentheses and are clustered at the county level. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table A3: Housing Equity and Not in Labor Force: Heterogeneity Effects

Panel A	White			Non-white		
	Total	Female	Male	Total	Female	Male
	(1)	(2)	(3)	(4)	(5)	(6)
Log Housing Equity	0.070***	0.089**	0.050	0.103	0.204	-0.007
	(0.027)	(0.038)	(0.032)	(0.079)	(0.130)	(0.062)
N	129000	67000	61500	17000	9700	7500
Efficient F	22.56	21.84	22.82	6.586	4.913	7.896
Panel B	Metro			Non-metro		
	Total	Female	Male	Total	Female	Male
Log Housing Equity	0.066**	0.099**	0.029	0.039	0.013	0.088
	(0.030)	(0.041)	(0.030)	(0.074)	(0.107)	(0.078)
N	116000	61500	55000	30000	15500	14000
Efficient F	18.74	17.27	20	8.023	7.229	7.966
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Ind and HH Controls	Yes	Yes	Yes	Yes	Yes	Yes
Local Controls	Yes	Yes	Yes	Yes	Yes	Yes
State Trends	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The sample is constructed from the restricted-use versions of the 1991–2004 SIPP panels, limited to individuals aged 15–55 who are homeowners, with observations covering the 1991–2005 period. The IV is constructed as the measure of households' duration of exposure to housing market booms driven by credit expansion in housing supply-constrained areas. Standard errors are presented in parentheses and are clustered at the county level. *** p < 0.01, ** p < 0.05, * p < 0.10.